Form FHWA-201 (Rev. 11-87)	U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION	Use this form in lieu of transmittal slips within Dept. of Trans. when message comment is to be retained as file material. Do not prepare carbons. Not to be used in lieu of Form FHWA-121 for informal correspondence.
	MINUTE - MEMO	

SUBJECT: NY ERFO WARI 2019-1(1) Scour Hydraulic Memo

TO	MESSAGE/COMMENT	FROM/DATE
Gwendolyn Riffle Project Manager	We have completed the scour analysis for the stream stabilization along the Winding Waters Trail within the Wallkill River National Wildlife Refuge in Warwick, New York. This memo has been revised to reflect changes to the scope and limits of work identified during the Plan-in-Hand site visit in mid-July 2025.	Isis Rosado-Vazquez Hydraulics Senior Technical Specialist
Canisha Lyons Highway Design Manager	The project scope includes streambank repair work at three locations. The streambank failures along the Winding Waters Trail (Sites 240 and Site 324) will be repaired using buried vegetated riprap revetment. At Site 3, an eroded bank at the outlet of a gully will be repaired using standard riprap revetment. All project sites are located within FEMA Zone AE mapped floodplains with designated Base Flood Elevations (BFEs).	ISIS Y ROSADO-VAZQUEZ Digitally signed by ISIS Y ROSADO-VAZQUEZ Date: 2025.12.05 14:33:12 -05'00' THROUGH Wendy Hiciano Hydraulics Team Leader
	<u>Winding Waters 240 (41.295016700, -74.526450000)</u> **<u>Winding Waters 324 (41.295733333, -74.525733333)</u>** Streambank failures along the Winding Waters Trail (Sites 240 and 324) will be repaired using stream-stabilization measures by realigning the trail and regrading the banks to create a flatter and more stable slopes suitable for installing buried vegetated riprap revetment as a scour countermeasure. Based on Chapter 7 of the PDDM, longitudinal embankments should be designed to withstand the estimated worst-case longitudinal scour up through the 100-year flood. The hydrologic analysis was performed using the USGS Streamstats® online automated hydrologic module which automates drainage area delineation from digital terrain models and computes peak flow discharges following standard regional regression equations. In addition to the main channel (Wallkill River), two unnamed tributaries were also considered in our hydrology analysis. Please refer to Table 1 for a summary of flow rates, and Table 2 and Table 3 for a summary of average water surfaces elevations (WSELs). Refer to Attachment 02 for StreamStats Reports; Attachment 03 and Attachment 04 for Site 240 SMS WSELs Plots; and Attachment 08 and Attachment 09 for Site 324 SMS WSELs Plots.	WENDY MERCEDES HICIANO Digitally signed by WENDY MERCEDES HICIANO Date: 2025.12.05 14:36:38 -05'00' 12/05/2025

FHWA: I. Rosado (6257): JM: M:\PROJECTS_FWS\WARI\NY ERFO FW WARI 2019-1(1)\ Techserv\ Hydra\ Reports\ Rev 03 - Riprap\ NY ERFO WARI 2019-1(1)_ Scour Memo 2025.12.05.docx
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Hydraulic modeling was conducted using the two-dimensional software Sedimentation and River Hydraulics – Two Dimensional (SMS SRH-2D) v13.3. The results indicate that the proposed streambank grading, which combines excavation with minor areas of fill, will not increase water surface elevations (WSELs) for any design-year flood event.

While minor filling within FEMA floodplains is generally discouraged, site constraints—including steep existing streambanks, wetlands across the trail, the minimum slope required for riprap installation, and the need to avoid channel encroachment—limit the extent of feasible excavation. The proposed grading balances these constraints while maintaining existing WSELs, and SMS results confirm that excavation exceeds the minor filling at eroded locations; as a result, the channel's cross-sectional area will increase rather than decrease, providing equal or greater conveyance than the original stream segment.

Sites 240 and 324 are located within FEMA Zone AE mapped floodplains with a Base Flood Elevation (BFE) of 388 ft, reflecting the relatively flat topography in the area (see Attachment 20 – FEMA FIRM# 36071C0417E). Results presented in Tables 2 and 3 indicate that existing and proposed WSELs remain 388.34 ft at Site 240 and 388.31 ft at Site 324. Attachment 21 further illustrates an SMS plan view comparing existing and proposed Q100 WSELs across the study area and project site, confirming that the proposed grading will not increase WSELs or cause physical damage to adjacent or downstream properties. No new structures requiring floodproofing are proposed.

Table 1: Hydrology Summary

Event (year)	Wallkill River Flow (cfs)	Upstream Tributary Flow (cfs)	Downstream Tributary Flow (cfs)
2	2030	46	44
10	3960	108	101
25	5220	150	140
50	6280	187	174
100	**7470**	**226**	**210**
200	8810	254	236
500	10800	338	313

Table 2: Site 240 Water Surface Elevation Summary

Flood-Event (year)	EX WSEL (ft)	PRO WSEL (ft)
2	381.53	381.46
10	384.46	384.46
25	386.11	386.11
50	387.22	387.21
100	388.34	388.34
200	389.64	389.63
500	391.35	391.34

Table 3: Site 324 Water Surface Elevation Summary

Flood-Event (year)	EX WSEL (ft)	PRO WSEL (ft)
2	381.22	381.10
10	384.24	384.24
25	386.05	386.05
50	387.19	387.18
100	388.32	388.31
200	389.63	389.62
500	391.34	391.33

Scour Analysis:

Scour analysis was performed according to HEC 23 V1, Section 4.3.5, Scour at Protected Bendways, using a (W/Dmnc) minimum value of 20, Rc/W of 1.5, and a Factor of Safety of 1.1. Riprap rock size was designed using the channel revetment riprap calculator within the FHWA HYD Toolbox V5.1 software, which is based on DG 4 of HEC 23 V2. The design parameters such as flow depth and flow velocities were obtained from two-dimensional software Sedimentation and River Hydraulics – Two Dimensional (SRH-2D) v13.3.

Table 4: Velocities Summary

	Channel Design Velocities	
	Average (fps)	Max (fps)
Site 204	4.45	5.44
Site 324	4.30	7.93

The D50 of 0.0002952756 ft. (0.09 mm) was obtained from boring logs samples at the sites, and subsequent particle size analysis performed for this location. Due to the visible presence of bedrock in the channel, which has been confirmed by borings at the site, long term degradation is considered negligible.

Site 240 - Based on the analysis, for the worst-bendway case total scour elevation of 365.70 ft was calculated from a thalweg elevation of 370.86 ft. at the bendway section. The average upstream flow depth in the main channel is 13.32 ft and the maximum water depth in the bend is 16.95ft. Based on the Geotechnical Report, bedrock elevation was assumed to range from 368.38 ft to 370.08 ft and the soil in the area is classified as a combination of SM (silty sand) and ML (low plasticity silt). Please refer to Table 5 for a summary of the computed scour elevations; the *Scour recommendation section* and Table 6 for final recommended scour elevations.

Site 324 - Based on the analysis, for the worst-bendway case total scour elevation of 363.74 ft was calculated from a thalweg elevation of 367.07 ft. at the bendway section. The average upstream flow depth in the main channel is 14.90 ft and the maximum water depth in the bend is 21.16ft. Based on the Geotechnical Report, the closest boring to this site, place the bedrock at elevation 370.08 ft and the soil in the area is classified as a combination of SM (silty sand) and ML (low plasticity silt). Please refer to Table 5 for a summary of the computed scour elevations; the *Scour recommendation section* and Table 6 for the HYD team's final recommended scour elevations.

Table 5: Computed Scour Depth Summary*

Event	Location	Scour Type	Scour Depth* (ft)	Thalweg Elevation at Bendway **(ft)	Scour Elevation (ft)
100 year (Design)	Site 240	Bendway scour depth with Factor of Safety	5.16		
		Long Term	0		
		TOTAL	**5.16**	**370.86**	**365.70*** **
	Site 324	Bendway scour depth with Factor of Safety	3.33		
		Long Term	0		
		TOTAL	**3.33**	**367.07**	**363.74*** **

Notes:

* See Table 6 for final recommended scour elevations.

** Thalweg is the required elevation for the bottom placement of revetment per FHWA guidelines.

*** Scour elevation is expected to be below the bedrock elevation.

Scour Recommendation:

Given that bedrock is visible in the channel, the borings confirm shallow bedrock, and the EFL HD and Construction team desire to minimize channel disturbance and avoid significant channel narrowing during construction operations, HYD will proceed as directed by the design team and will run the scour analysis setting the bottom of the proposed revetment at the scour depth specified in Table 6, Attachment 07 (Site 240) and Attachment 13 (Site 324) toe layout, instead of the HEC 23V2 Figure 4.2 or Figure 4.3 toe layout.

Based on the proposed contours and alignment provided by HD, the elevation of the toe of embankment (TOE) varies along the length of the proposed revetment. As a result, Table 6 reflects only the lowest TOE elevation along the proposed revetment.

Table 6: Recommended Scour Depth Summary

Event	Location	Scour Type	Scour Depth** (ft)	Lowest Toe elevation along proposed revetment* (ft)	Scour Elevation (ft)
100 year (Design)	Site 240	Bendway scour depth with Factor of Safety	5.16		
		Long Term	0		
		TOTAL	**5.16**	**373.54**	**368.38*** **
	Site 324	Bendway scour depth with Factor of Safety	3.33		
		Long Term	0		
		TOTAL	**3.33**	**370.91**	**367.58*** **

Notes:

* Reflects the lowest elevation along the TOE of the proposed revetment.

** Scour depth is referenced from the TOE elevation and not the thalweg elevation.

*** Scour elevation may be below the expected bedrock elevation.

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Velocities:

Banks:
No increase in water velocity along the banks is anticipated, as the proposed vegetated riprap will maintain the existing surface roughness. To address areas of highest hydraulic shear stress, the toe of the embankment will be protected with Class 5 exposed riprap revetment.

Above the ordinary high-water mark, Class 3 riprap will be buried within the bank and overlaid with native vegetation. This approach preserves surface roughness to sustain stable flow velocities while reinforcing the embankment toe against scour.

Toe of embankment (TOE):

Flow velocities along the proposed TOE for existing and proposed conditions were assessed for the Riprap Class 5 using the Observation Plot tool within the SMS software. Refer to Attachment 05 (Site 240) and Attachment 10 (Site 324) for SMS velocity plots.

Our analysis indicates that the increase in flow velocity downstream of the riprap treatment limits between the Class 5 toe riprap and the adjacent existing ground is less than 1 fps. This minor increase in velocity is not anticipated to adversely impact the performance or stability of the vegetated riprap revetment, the banks, or the surrounding channel given that the average velocities remain within the tolerable limits for typical bank materials and vegetation and are therefore considered to be within the design threshold for bank stability and erosion resistance.

Table 7: Site 240 - Velocities along the proposed Riprap TOE

Site 240	
	Velocity at the downstream edge of riprap (fps)
EX	2.41
PRO	2.03
Difference in Velocity due change in roughness along the TOE from Riprap Class 5 to existing ground	**- 0.38** (bedrock is exposed and visible downstream of the riprap limits of work)

Table 8: Site 324 - Velocities along the proposed Riprap TOE

Site 324	
	Velocity at the downstream edge of riprap (fps)
EX	2.0
PRO	3.8
Difference in Velocity due change in roughness along the TOE from Riprap Class 5 to existing ground	+ **0.8**

Bioengineering analysis:

Vegetated Riprap revetment recommendation was performed based on HEC 23 V1, Chapter 6, Biotechnical Engineering. USDA - Chapter 16 –Soil Bioengineering for Streambank and Shoreline Protection, 2021, was also considered in this analysis. Riprap rock size was designed using the channel revetment riprap calculator within the FHWA HYD Toolbox V5.1 software, which is based on HEC 23 V2, DG 4, Riprap Revetment. Large riprap at the toe of embankment was calculated using a factor of safety of 1.5. Refer to Attachment 06 (Site 240) and Attachment 12 (Site 324) for Hyd Toolbox Reports.

The bioengineering alternatives evaluated during our analysis are the following:

Table 9: Bioengineering Design Criteria and Recommendation Summary

			Site 240		**Site 324**	
Practice	Permissible shear stress* (lb/ft^2)	Permissible Velocity* (ft/s)	Shear stress (lb/ft^2)	Velocity (ft/s)	Shear stress (lb/ft^2)	Velocity (ft/s)
Live stakes/poles	Initial: 0.5 to 2 Established: 2 to 5+	Initial: 1 to 2.5+ Established: 3 to 10	0.30 (avg) 0.56 (max)	4.45 (avg) 5.44(max)	2.29 (avg) 5.40 (max)	4.30 (avg) 7.93 (max)
Live stakes/poles and riprap	Initial: 3+ Established: 6 to 8+	Initial: 5 to 10+ Established: 12+				
Turf Reinforcement Mats	Established: 6+ (5.A, FP-24 Table 713.4)	Established: 3 to 8				
		HYD Recommendation:	**Live stakes/poles with buried riprap**		**Turf Reinforcement Mat with buried riprap or Live stakes/poles with buried riprap**	

Notes:

* Figure 16-19. Design Criteria for various Streambank Soil Bioengineering Practices, USDA – Chapter 16 –Soil Bioengineering for Streambank and Shoreline Protection, 2021.

** Based on our analysis, floodplain overtopping across the trail at Site 324 is causing higher flow velocities and shear stress along the banks compared to Site 240.

*** To provide a more conservative basis for design, the HYD Toolbox values were used for design purposes, as they were higher than those obtained from SMS.

**** To provide a more conservative basis for design, the values reflect the higher velocity and shear stresses observed within the channel cross-section at each respective site, not only the eroded banks.

Based on our analysis, vegetated riprap in a form of a combination of live stakes/pole plantings and riprap is recommended for **Site 240**, as the calculated shear stress and velocity values fall within the permissible limits for live stakes/poles once they are fully established. Refer to Attachment 07 for the draft typical section of live pole plantings and riprap.

Site 324 is expected to experience higher flow velocities and shear stress than Site 240, along with a greater risk of erosion from overbank overtopping. As such, this memo presents two design alternatives for Site 324:

Alternative 01: Turf Reinforcement Mat with buried riprap (see Attachment 13a for the typical section)
Alternative 02: Live stakes/pole plantings with buried riprap (see Attachment 13b for the typical section)

While live pole planting with riprap at Site 324 falls within permissible hydraulic limits once fully established, the HYD team recommends using Alternative 01, turf reinforcement with buried riprap, as a more conservative design approach. Turf reinforcement mats offer greater resistance to shear stress compared to live stakes or poles, providing added protection under higher flow conditions.

Table 10. Site 240 and Site 324 Riprap Summary

Location	Riprap Type	Riprap Class	D50 (in)	Riprap Thickness (in)	Maximum Slope	Length (ft)*	Width (ft)	Bedding filter thickness (in)	Geotextile
Site 240 & Site 324	Angular	3 along Bank	12	24	1V:1.5H	+/- 158	Varies	6	Yes
		5 along TOE	18	54**	1V:1.5H	+/- 226	Varies	9**	No**

Notes:
* HEC-23 recommends that riprap along bendway extend one channel width upstream of the Point of Curvature (PC) and 1.5 channel widths downstream of the Point of Tangency (PT). The riprap limits in this project do not meet this FHWA requirement. During the Plan-in-Hand site visit in mid-July 2025, the length of the vegetated riprap was reduced to minimize tree removal.
** To provide mitigation for installation below the ordinary high-water mark.

Vegetated riprap includes the installation of pole plantings, void-filling with appropriate materials, topsoil placement, and measures for vegetation establishment. Continued inspection and maintenance will be essential to ensure the long-term performance and stability of the bioengineered banks. Refer to Attachment 07 (Site 240) and Attachment 13 (Site 324) for site-specific draft vegetated riprap details.

The recommended spacing for pole plantings is 3 feet. Container plant spacing will vary based on the proposed planting plan and is subject to approval by the CO. For design purposes, a spacing of 4.5 feet is recommended as a general assumption.

Temporary Conditions:

Based on Construction's feedback, stream diversion is not necessary because the work will be performed during the low-flow season. Hydraulics estimates that the low-flow period extends from June through August.

Winding Waters Site 3 (41.294566670, -74.529000000)

Based on the DSR, the scope of work is to address the bank erosion at the end of a gully. Hydraulic modeling was performed using the two-dimensional software Sedimentation and River Hydraulics – Two-Dimensional (SRH-2D, SMS) v13.3. Riprap was designed per Hydraulic Engineering Circular (HEC) 23, Volume 2, Guideline 4 using the Hydraulic Toolbox V4.4.

Based on the analysis, the main channel average upstream flow depth is 15.19 ft, and the average velocity is 2.38 ft/s. Please refer to Table 12 for a summary of flow rates with their corresponding WSELs, Attachment 14 and Attachment 15 for SMS WSELs Plots and Attachment 16 for the SMS Velocity Plot.

Table 12: Site 3 Water Surface Elevation Summary

Event (year)	EX WSEL (ft)	PRO WSEL (ft)
2	382.21	382.18
10	385.19	385.14
25	386.51	386.48
50	387.52	387.51
100	388.55	388.55
200	389.78	389.78
500	391.44	391.44

We recommend placing a Class 3 riprap revetment with a minimum depth of 24-inch at a maximum slope of 1V:1.5H up to the top of the bank to inhibit the risk of future erosion. Refer to Attachment 19 for the riprap plan view layout and proposed contours; Attachment 18 for Cross Section; and Attachment 17 for Riprap Analysis (HYD Toolbox Report).

Table 13. Riprap Summary

Location	Riprap Type	Riprap Class	D50 (in)	Riprap Thickness (in)	Maximum Slope	Length (ft)	Width (ft)
Site 3	Angular	3	12	30	1V:1.5H	+/- 26	6

Please contact us if you should have any questions or concerns.

ATTACHMENTS

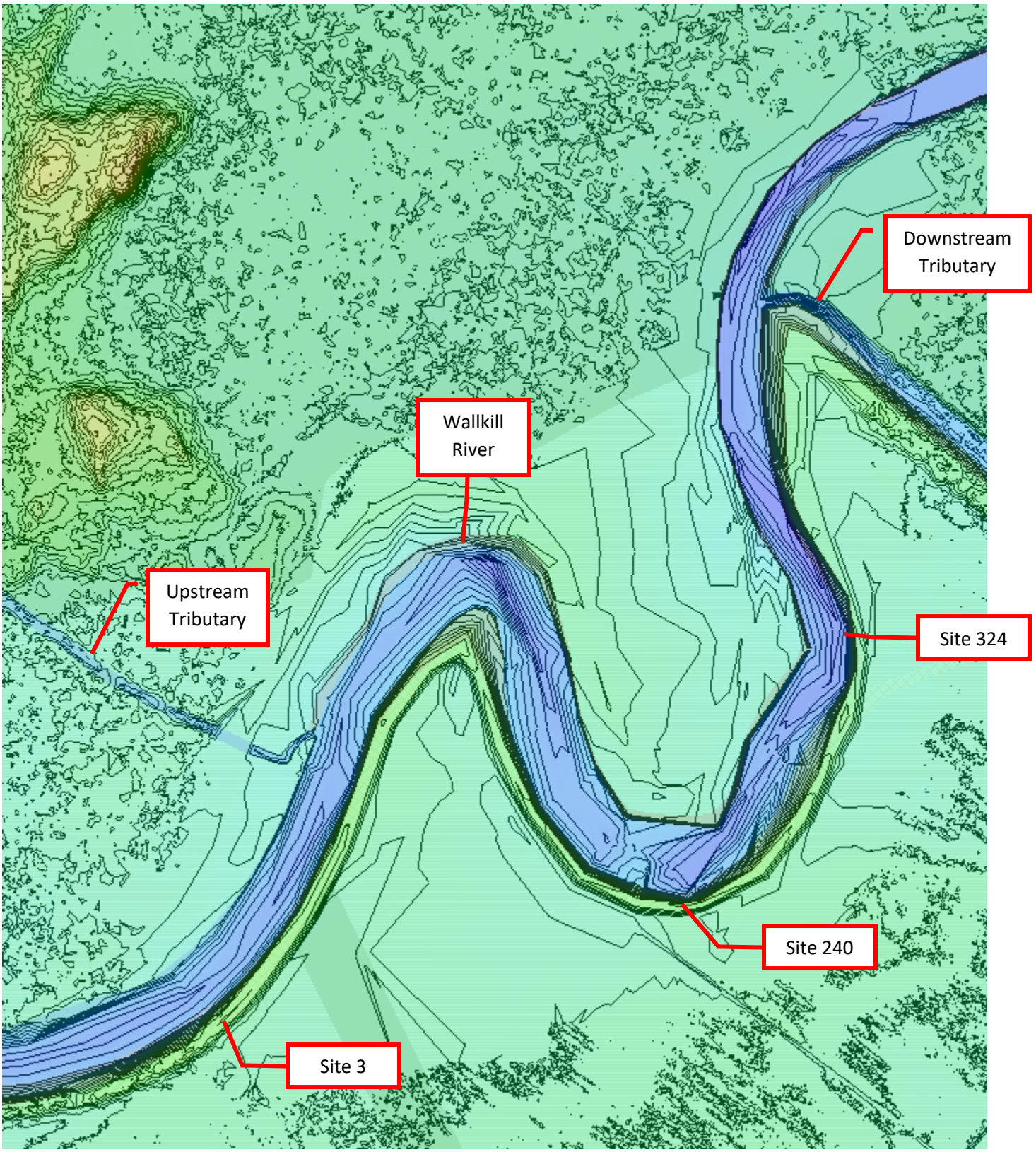
The following attachments can be found in the following link:
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- Attachment 01 - NY WARI 2019-1(1) - SMS Plan View.pdf
- Attachment 02 - NY WARI 2019-1(1) - Hydrology Streamstats.pdf
- Attachment 03 - NY WARI 2019-1(1) - Site 240 EX Wsels.pdf
- Attachment 04 - NY WARI 2019-1(1) - Site 240 PRO Wsels.pdf
- Attachment 05 - NY WARI 2019-1(1) - Site 240-Velocities.pdf
- Attachment 06 - NY WARI 2019-1(1) - Riprap Toolbox - Site 240 .pdf
- Attachment 07 - E251-A_Vegetated Riprap - Site 240.pdf
- Attachment 08 - NY WARI 2019-1(1) - Site 324 EX Wsels.pdf
- Attachment 09 - NY WARI 2019-1(1) - Site 324 PRO Wsels.pdf
- Attachment 10 - NY WARI 2019-1(1) - Site 324-Velocities.pdf
- Attachment 11 - NY WARI 2019-1(1) - Site 324-Bank Shear Stress.pdf
- Attachment 12 - NY WARI 2019-1(1) - Riprap Toolbox - Site 324 .pdf
- Attachment 13a - E251-B_Vegetated Riprap- Site 324.pdf
- Attachment 13b - E251-B_Vegetated Riprap- Site 324.pdf
- Attachment 14 - NY WARI 2019-1(1) - Site 3 EX WSEL's Plot.pdf
- Attachment 15 - NY WARI 2019-1(1) - Site 3 PRO WSEL's Plot.pdf
- Attachment 16 - NY WARI 2019-1(1) - Site 3 Velocities.pdf
- Attachment 17 - NY WARI 2019-1(1) - Site 3 Riprap Toolbox.pdf
- Attachment 18 - NY WARI 2019-1(1) - Site 3 Cross Section.pdf
- Attachment 19 - NY WARI 2019-1(1) - Site 3 Plan View.pdf
- Attachment 20 - NY WARI 2019-1(1) - FEMA FIRM.pdf
- Attachment 21 - NY WARI 2019-1(1) - SMS Flooplain WSEL Comparison.pdf
- E251-A_Buried Riprap - Underwater.dgn
- Section 251 SCR

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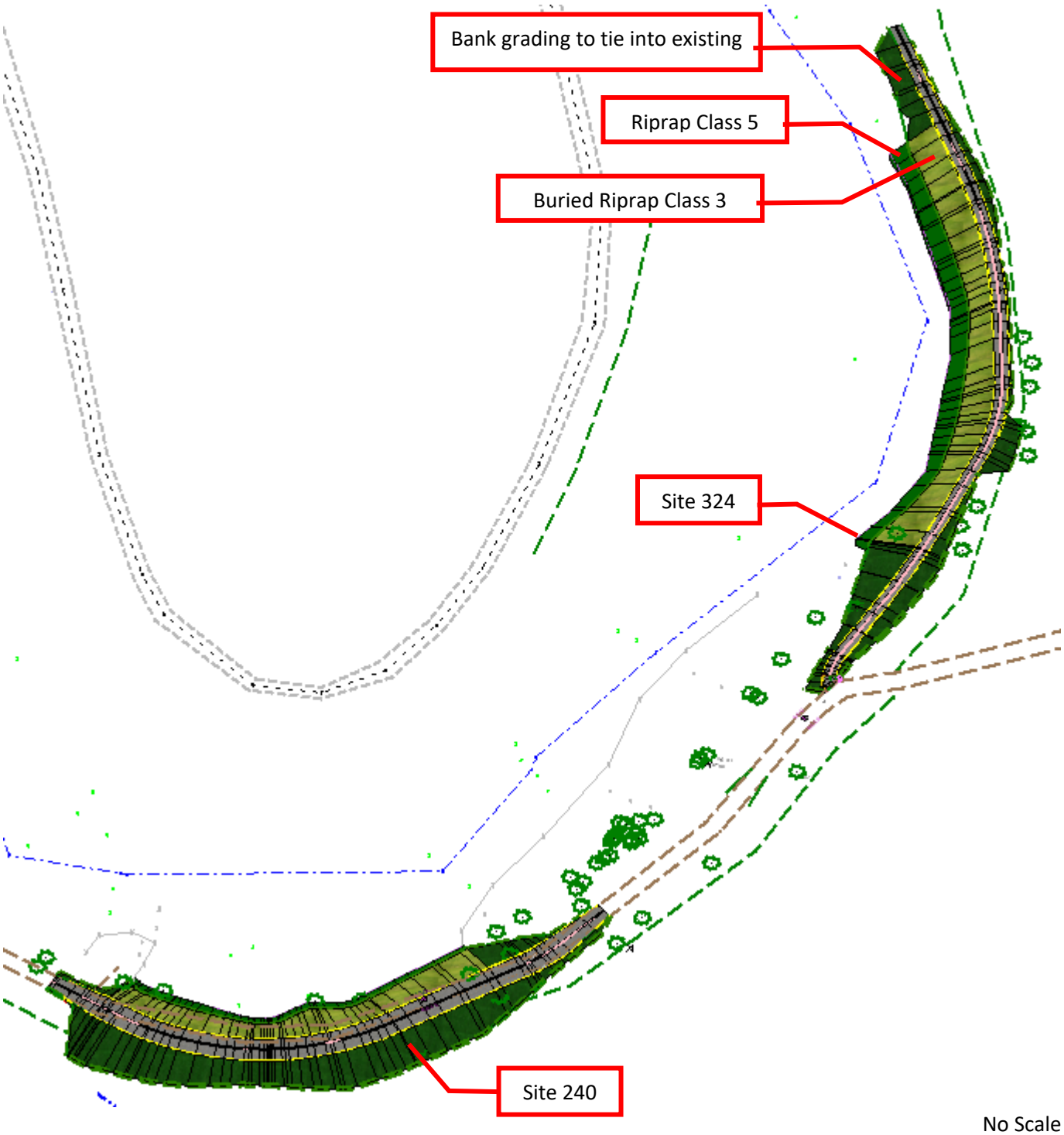
Attachment 1: NY WARI 2019-1(1)

SMS Plan View



No Scale

Vegetated riprap plan view in ORD — intended for visualization only

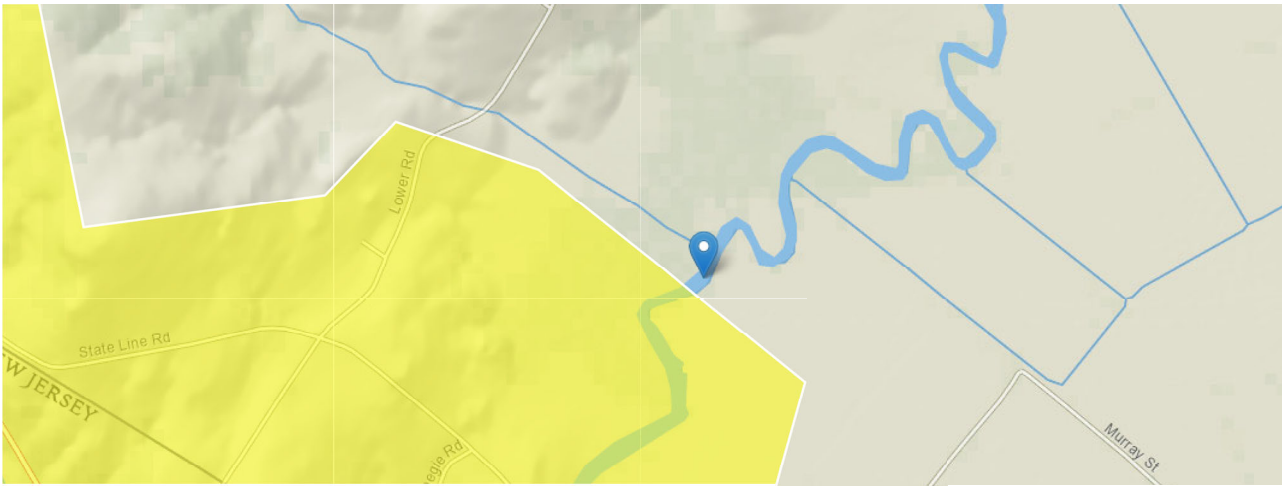


No Scale

Attachment 02a: Wallkill River

StreamStats Report- Stream Stabilization SMS Inflow BC

Region ID: NY
Workspace ID: NY20220512202205943000
Clicked Point (Latitude, Longitude): 41.29483, -74.52905
Time: 2022-05-12 16:22:32 -0400



Basin Characteristics

Parameter Code	Parameter Description	Value	Unit
BSLOPCM	Mean basin slope determined by summing lengths of all contours in basin mulitplying by contour interval and dividing product by drainage area	552	feet per mi
CENTROIDX	Basin centroid horizontal (x) location in state plane coordinates	532368.4	meters
CENTROIDY	Basin centroid vertical (y) location in state plane units	4558841.3	meters
CONTOUR	Total length of all elevation contours in drainage area in miles	845.8	miles
CSL1085LO	10-85 slope of lower half of main channel in feet per mile.	2.67	feet per mi
CSL1085UP	10-85 slope of upper half of main channel in feet per mile.	18.7	feet per mi
CSL10_85	Change in elevation divided by length between points 10 and 85 percent of distance along main channel to basin divide - main channel method not known	13.6	feet per mi
DRNAREA	Area that drains to a point on a stream	153	square miles
EL1200	Percentage of basin at or above 1200 ft elevation	6.08	percent
FOREST	Percentage of area covered by forest	53.4	percent
JULAVPRE	Mean July Precipitation	4.31	inches
JUNAVPRE	Mean June Precipitation	4.53	inches
JUNMAXTMP	Maximum June Temperature, in degrees F	77.7	degrees F
LAGFACTOR	Lag Factor as defined in SIR 2006-5112	3.7	dimensionless
LC11DEV	Percentage of developed (urban) land from NLCD 2011 classes 21-24	13.1	percent
LC11IMP	Average percentage of impervious area determined from NLCD 2011 impervious dataset	2.04	percent
LENGTH	Length along the main channel from the measuring location extended to the basin divide	31.5	miles
MAR	Mean annual runoff for the period of record in inches	22.3	inches
MAYAVPRE	Mean May Precipitation	4.55	inches
MXSNO	50th percentile of seasonal maximum snow depth from Northeast Regional Climate Center atlas by Cember and Wilks, 1993	11.2	inches
OUTLETX	Basin outlet horizontal (x) location in state plane coordinates	539435	feet
OUTLETY	Basin outlet vertical (y) location in state plane coordinates	4571595	feet
PRECIP	Mean Annual Precipitation	45.1	inches

Parameter Code	Parameter Description	Value	Unit
PRJUNAUG00	Basin average mean precip for June to August from PRISM 1971-2000	13	inches
SLOPERATIO	Ratio of main channel slope to basin slope as defined in SIR 2006-5112	0.0246	dimensionless
SSURGOA	Percentage of area of Hydrologic Soil Type A from SSURGO	13.3	percent
SSURGOB	Percentage of area of Hydrologic Soil Type B from SSURGO	16.6	percent
STORAGE	Percentage of area of storage (lakes ponds reservoirs wetlands)	6.98	percent

Peak-Flow Statistics Parameters [99.7 Percent (153 square miles) 2006 Full Region 2]

Parameter Code	Parameter Name	Value	Units	Min Limit	Max Limit
DRNAREA	Drainage Area	153	square miles	1.93	996
LAGFACTOR	Lag Factor	3.7	dimensionless	0.014	6.997
STORAGE	Percent Storage	6.98	percent	0	11.88
MAR	Mean Annual Runoff in inches	22.3	inches	16.03	33.95

Peak-Flow Statistics Flow Report [99.7 Percent (153 square miles) 2006 Full Region 2]

PII: Prediction Interval-Lower, Plu: Prediction Interval-Upper, ASEp: Average Standard Error of Prediction, SE: Standard Error (other -- see report)

Statistic	Value	Unit	SE	ASEp	Equiv. Yrs.
80-percent AEP flood	1400	ft^3/s	25.5	25.5	4.8
66.7-percent AEP flood	1660	ft^3/s	25.6	25.6	4.3
50-percent AEP flood	2030	ft^3/s	25.8	25.8	4.4
20-percent AEP flood	3100	ft^3/s	27	27	7.3
10-percent AEP flood	3960	ft^3/s	28.2	28.2	10.1
4-percent AEP flood	5220	ft^3/s	29.9	29.9	13.6
2-percent AEP flood	6280	ft^3/s	31.5	31.5	15.8
1-percent AEP flood	7470	ft^3/s	33.3	33.3	17.6
0.5-percent AEP flood	8810	ft^3/s	35.3	35.3	18.9
0.2-percent AEP flood	10800	ft^3/s	38.4	38.4	20.1

Peak-Flow Statistics Citations

Lumia, Richard, Freehafer, D.A., and Smith, M.J.,2006, Magnitude and Frequency of Floods in New York: U.S. Geological Survey Scientific Investigations Report 2006–5112, 152 p. (http://pubs.usgs.gov/sir/2006/5112/)

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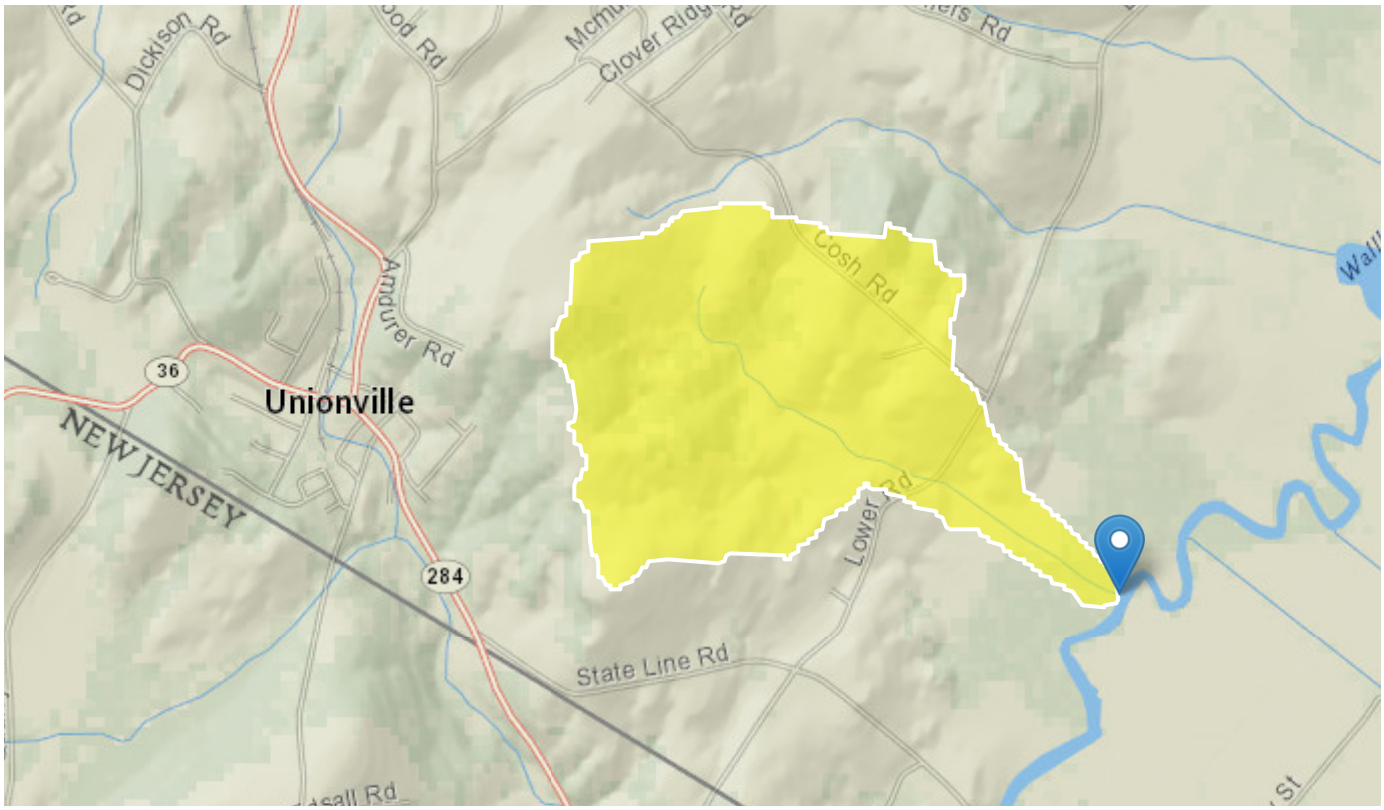
USGS Product Names Disclaimer: Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

Application Version: 4.8.1
StreamStats Services Version: 1.2.22
NSS Services Version: 2.1.2

Attachment 02b: Upstream Tributary

Tributary 1 - Winding Waters StreamStats Report

Region ID: NY
Workspace ID: NY20220412140837950000
Clicked Point (Latitude, Longitude): 41.29564, -74.52885
Time: 2022-04-12 10:09:06 -0400



Basin Characteristics

Parameter Code	Parameter Description	Value	Unit
DRNAREA	Area that drains to a point on a stream	0.66	square miles
LAGFACTOR	Lag Factor as defined in SIR 2006-5112	0.00783	dimensionless
MAR	Mean annual runoff for the period of record in inches	20.3	inches
STORAGE	Percentage of area of storage (lakes ponds reservoirs wetlands)	0.0175	percent

Peak-Flow Statistics Parameters [2006 Full Region 2]

Parameter Code	Parameter Name	Value	Units	Min Limit	Max Limit
DRNAREA	Drainage Area	0.66	square miles	1.93	996
LAGFACTOR	Lag Factor	0.00783	dimensionless	0.014	6.997
STORAGE	Percent Storage	0.0175	percent	0	11.88
MAR	Mean Annual Runoff in inches	20.3	inches	16.03	33.95

Peak-Flow Statistics Disclaimers [2006 Full Region 2]

One or more of the parameters is outside the suggested range. Estimates were extrapolated with unknown errors.

Peak-Flow Statistics Flow Report [2006 Full Region 2]

Statistic	Value	Unit
80-percent AEP flood	27.9	ft^3/s
66.7-percent AEP flood	35.2	ft^3/s
50-percent AEP flood	46	ft^3/s
20-percent AEP flood	79.9	ft^3/s
10-percent AEP flood	108	ft^3/s
4-percent AEP flood	150	ft^3/s
2-percent AEP flood	187	ft^3/s
1-percent AEP flood	226	ft^3/s
0.5-percent AEP flood	271	ft^3/s
0.2-percent AEP flood	338	ft^3/s

Peak-Flow Statistics Citations

Lumia, Richard, Freehafer, D.A., and Smith, M.J.,2006, Magnitude and Frequency of Floods in New York: U.S. Geological Survey Scientific Investigations Report 2006–5112, 152 p. (http://pubs.usgs.gov/sir/2006/5112/)

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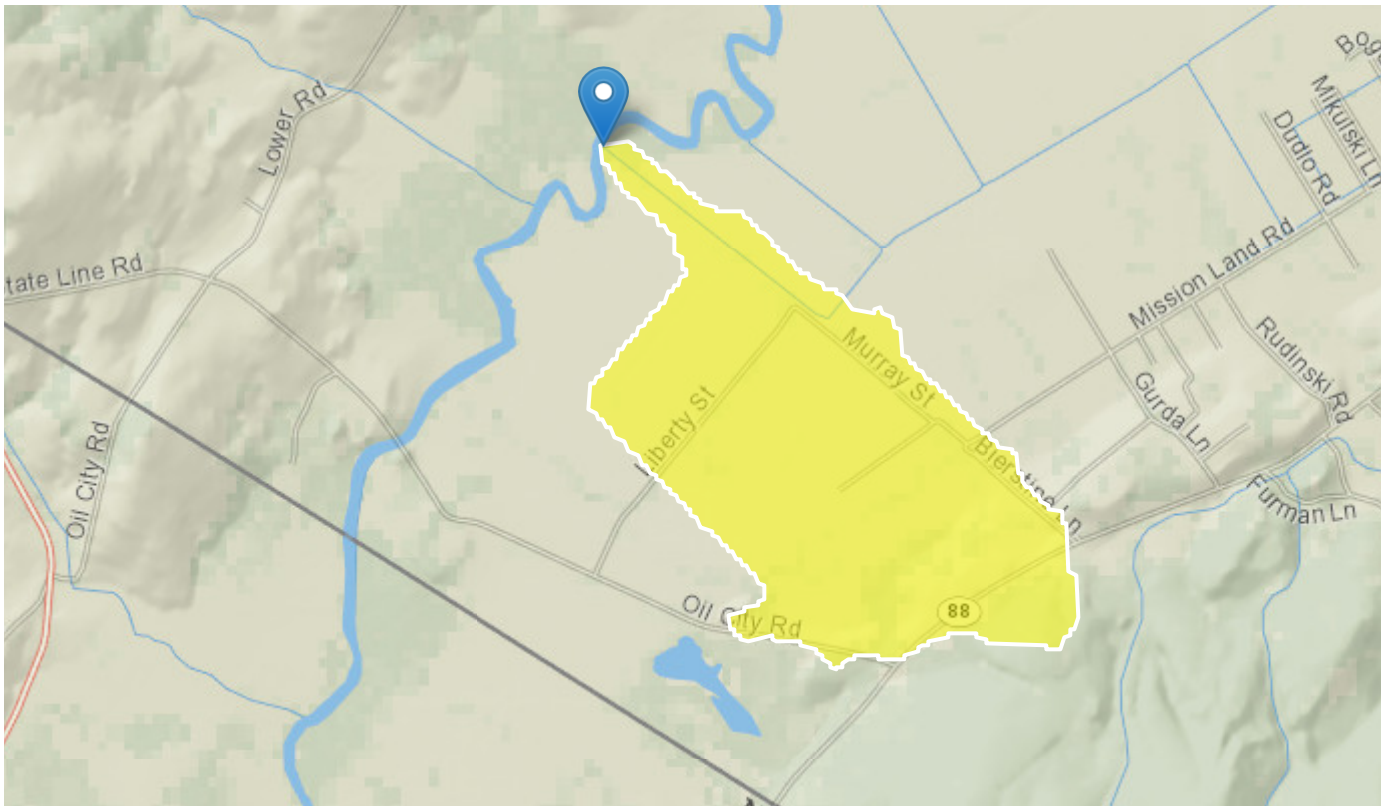
USGS Product Names Disclaimer: Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

Application Version: 4.8.1
StreamStats Services Version: 1.2.22
NSS Services Version: 2.1.2

Attachment 02c: Downstream Tributary

Tributary 2 - StreamStats Report

Region ID: NY
Workspace ID: NY20220412141357205000
Clicked Point (Latitude, Longitude): 41.29754, -74.52584
Time: 2022-04-12 10:14:25 -0400



Basin Characteristics

Parameter Code	Parameter Description	Value	Unit
CENTROIDX	Basin centroid horizontal (x) location in state plane coordinates	540501.8	meters
CENTROIDY	Basin centroid vertical (y) location in state plane units	4570804.5	meters
CSL1085LO	10-85 slope of lower half of main channel in feet per mile.	14.6	feet per mi
DRNAREA	Area that drains to a point on a stream	0.64	square miles
EL1200	Percentage of basin at or above 1200 ft elevation	0	percent

Parameter Code	Parameter Description	Value	Unit
JULAVPRE	Mean July Precipitation	4.05	inches
JUNAVPRE	Mean June Precipitation	4.37	inches
JUNMAXTMP	Maximum June Temperature, in degrees F	79	degrees F
LAGFACTOR	Lag Factor as defined in SIR 2006-5112	0.0428	dimensionless
LENGTH	Length along the main channel from the measuring location extended to the basin divide	1.85	miles
MAR	Mean annual runoff for the period of record in inches	19.7	inches
MAYAVPRE	Mean May Precipitation	4.36	inches
PRJUNAUG00	Basin average mean precip for June to August from PRISM 1971-2000	12.5	inches
SSURGOA	Percentage of area of Hydrologic Soil Type A from SSURGO	2.42	percent
SSURGOB	Percentage of area of Hydrologic Soil Type B from SSURGO	10.8	percent
STORAGE	Percentage of area of storage (lakes ponds reservoirs wetlands)	0	percent

Peak-Flow Statistics Parameters [2006 Full Region 2]

Parameter Code	Parameter Name	Value	Units	Min Limit	Max Limit
DRNAREA	Drainage Area	0.64	square miles	1.93	996
LAGFACTOR	Lag Factor	0.0428	dimensionless	0.014	6.997
STORAGE	Percent Storage	0	percent	0	11.88
MAR	Mean Annual Runoff in inches	19.7	inches	16.03	33.95

Peak-Flow Statistics Disclaimers [2006 Full Region 2]

One or more of the parameters is outside the suggested range. Estimates were extrapolated with unknown errors.

Peak-Flow Statistics Flow Report [2006 Full Region 2]

Statistic	Value	Unit
80-percent AEP flood	26.4	ft^3/s
66.7-percent AEP flood	33.3	ft^3/s
50-percent AEP flood	43.4	ft^3/s
20-percent AEP flood	74.9	ft^3/s
10-percent AEP flood	101	ft^3/s
4-percent AEP flood	140	ft^3/s
2-percent AEP flood	174	ft^3/s
1-percent AEP flood	210	ft^3/s
0.5-percent AEP flood	251	ft^3/s
0.2-percent AEP flood	313	ft^3/s

Peak-Flow Statistics Citations

Lumia, Richard, Freehafer, D.A., and Smith, M.J.,2006, Magnitude and Frequency of Floods in New York: U.S. Geological Survey Scientific Investigations Report 2006–5112, 152 p. (http://pubs.usgs.gov/sir/2006/5112/)

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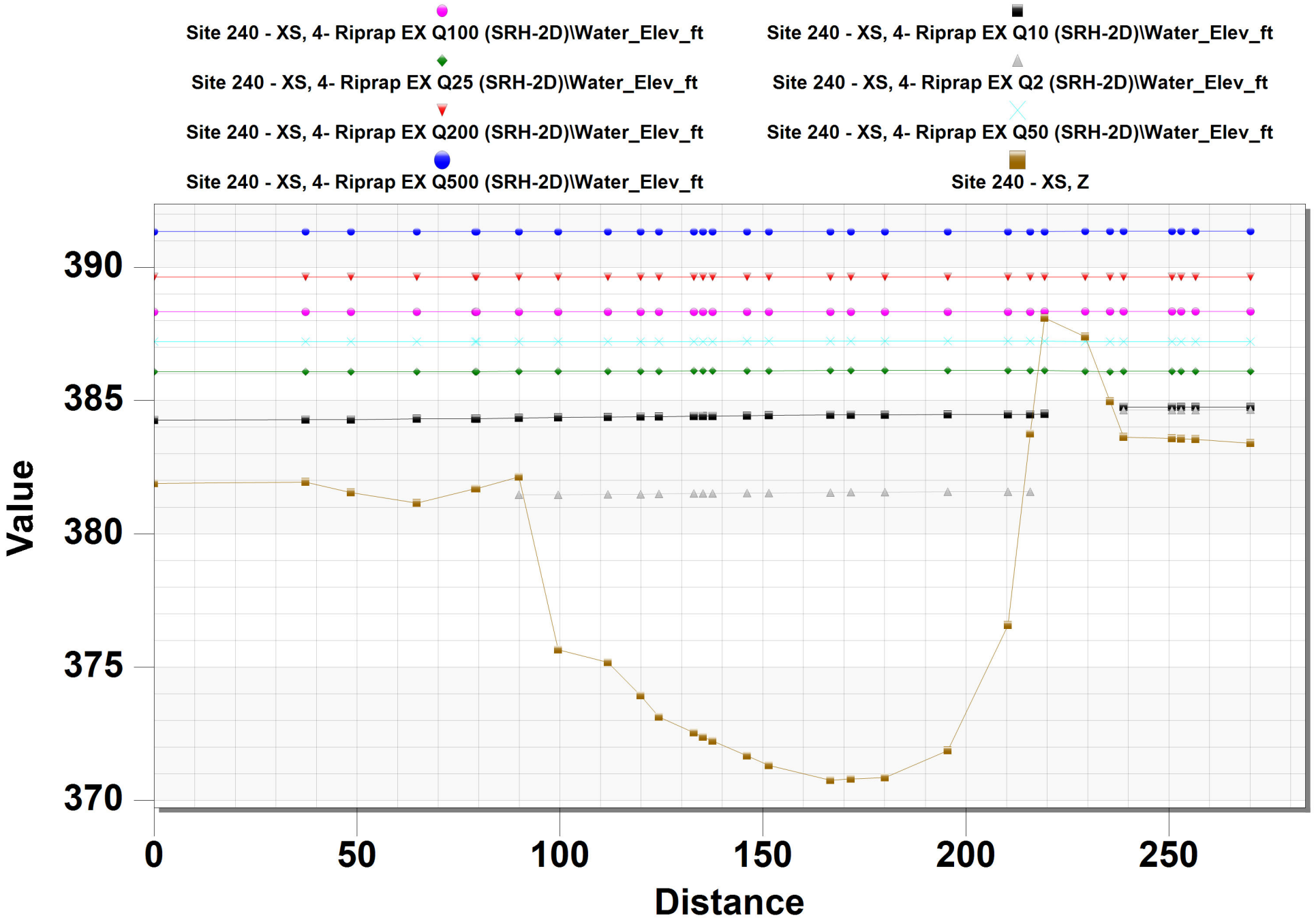
USGS Product Names Disclaimer: Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

Application Version: 4.8.1
StreamStats Services Version: 1.2.22
NSS Services Version: 2.1.2

Attachment 03:

NY WARI 2019-1(1)

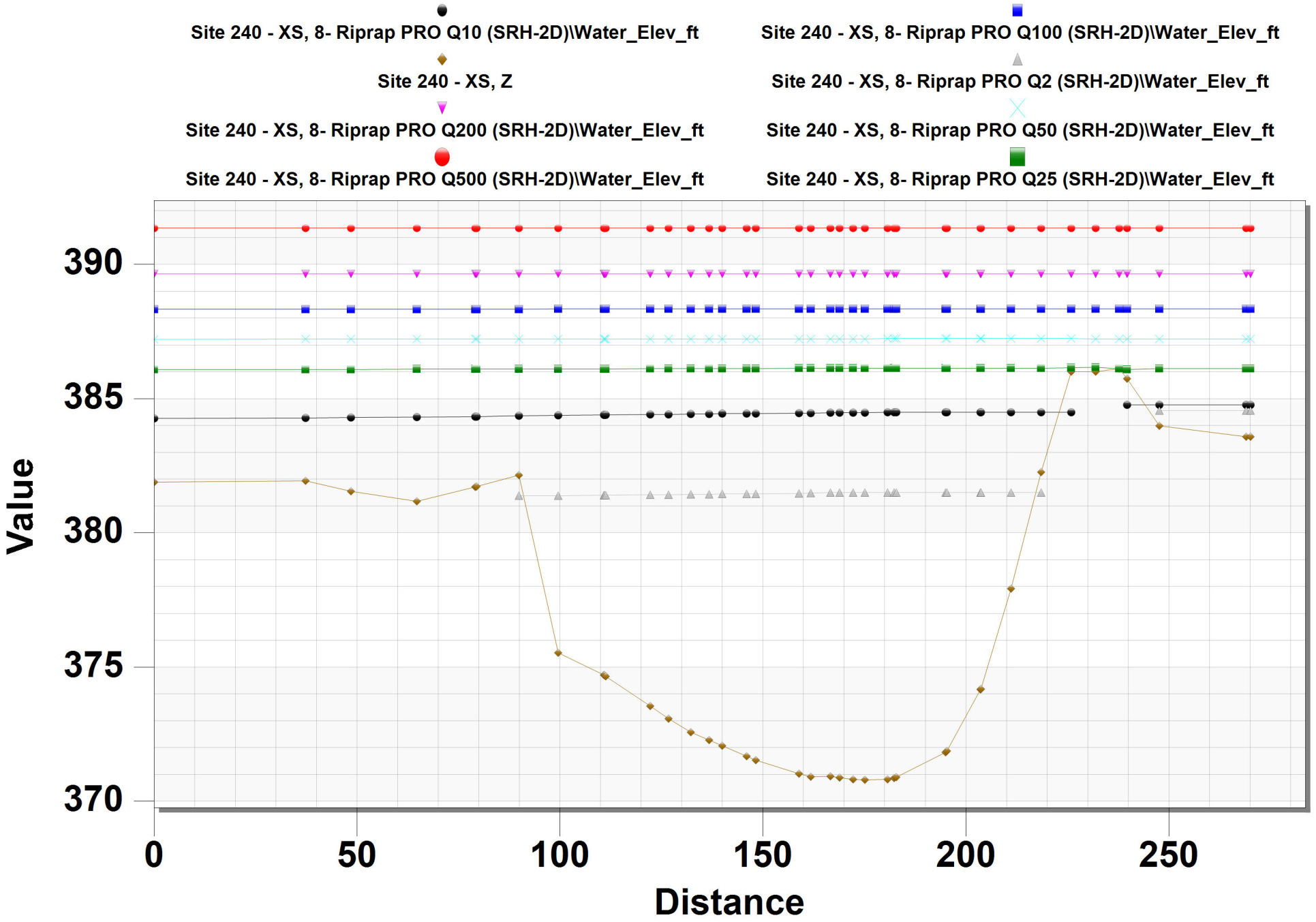
Site 240 - EX Wsels



Attachment 04:

NY WARI 2019-1(1)

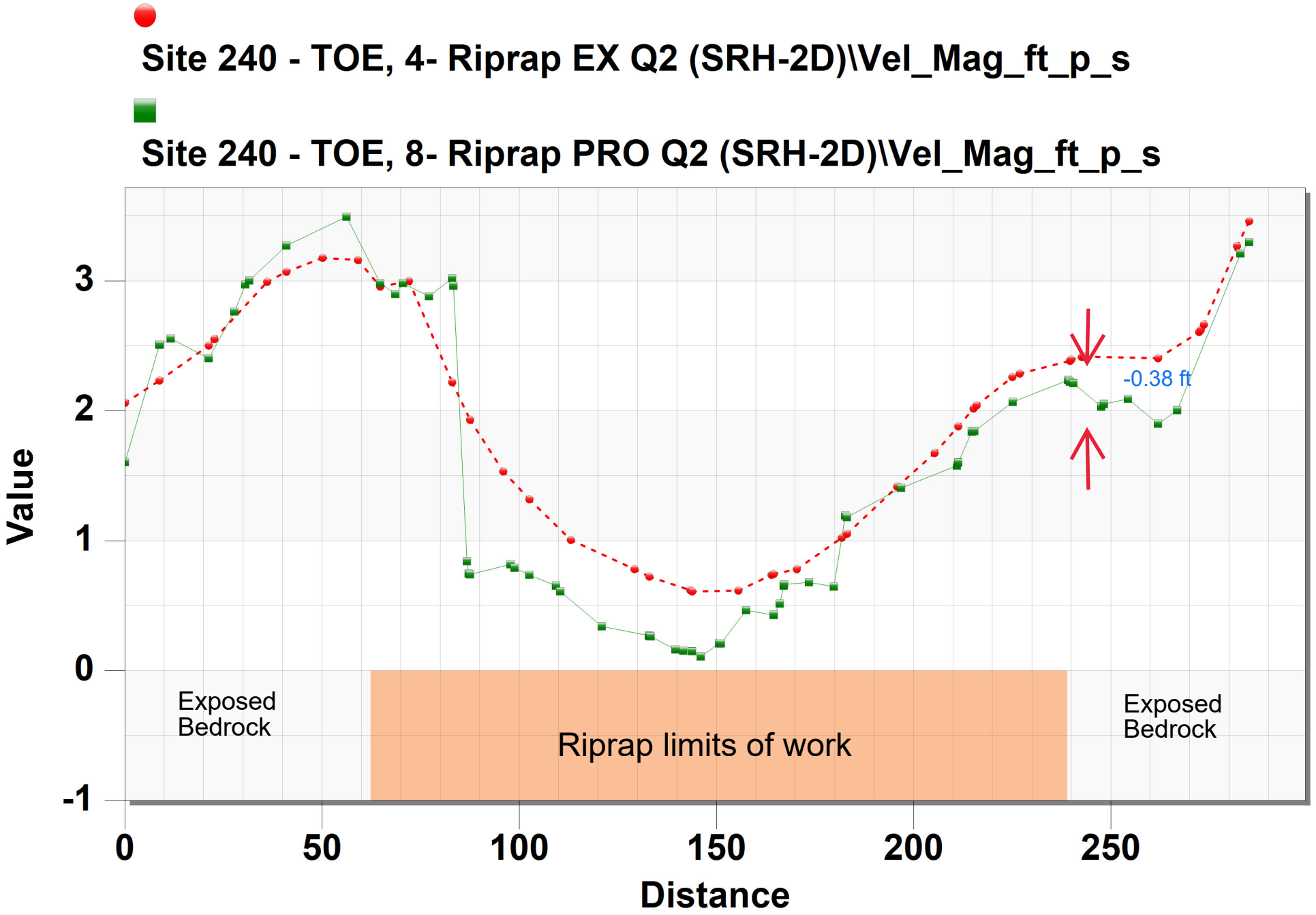
Site 240 - PRO Wsels



Attachment 05:

NY WARI 2019-1(1)

Site 240 - TOE - Velocities (fps)



Attachment 06:

Hydraulic Analysis Report- Site 240

Project Data

Project Title: NY WARI 2019-1(1)

Designer: EFL HYD (IRV)

Project Date: Thursday, December 04, 2025

Project Units: U.S. Customary Units

Notes: Q100

Channel Analysis: Channel Analysis 240 XS Q100

Notes:

Input Parameters

Channel Type: Custom Cross Section

Cross Section Data

Station (ft	Elevation (ft)	Manning's n
0.00	381.46	0.0350
2.28	381.39	0.0350
10.81	381.11	0.0350
15.67	381.18	0.0350
39.81	382.17	0.0350
42.96	382.30	0.0350
44.49	377.56	0.0350
48.03	375.65	0.0350
58.99	375.17	0.0350
68.34	375.02	0.0350
70.69	373.23	0.0350
74.94	372.83	0.0350
106.59	370.80	0.0350
128.75	370.52	0.0350
142.22	371.28	0.0350
150.58	373.00	0.0350
153.13	374.89	0.0350
159.70	376.62	0.0350
160.12	376.74	0.0350
160.41	376.81	0.0350
160.44	376.82	0.0350
160.48	376.83	0.0350
167.07	380.25	0.0350
181.35	387.46	0.0350
182.21	387.50	0.0350

183.35	387.54	0.0350
185.40	387.62	0.0350
187.36	387.70	0.0350
189.86	387.60	0.0350
191.37	387.54	0.0350
192.76	387.48	0.0350
193.38	387.46	0.0350
195.46	386.77	0.0350
198.54	385.74	0.0350
199.46	385.43	0.0350
207.40	383.53	0.0350
207.84	383.52	0.0350
209.18	383.51	0.0350
209.91	383.50	0.0350
215.89	383.44	0.0350
216.09	383.44	0.0350
219.20	383.41	-----

Longitudinal Slope: 0.0005 ft/ft

Depth 16.9500 ft

Result Parameters

Flow 9742.7300 cfs

Area of Flow 2170.9593 ft^2

Wetted Perimeter 226.3406 ft

Hydraulic Radius 9.5916 ft

Average Velocity 4.4878 ft/s

Top Width 207.4394 ft

Froude Number: 0.2445

Critical Depth 8.2034 ft

Critical Velocity 13.7757 ft/s

Critical Slope: 0.0103 ft/ft

Critical Top Width 120.00 ft

Calculated Max Shear Stress 0.5458 lb/ft^2

Calculated Avg Shear Stress 0.3088 lb/ft^2

Composite Manning's n Equation: Lotter method

Manning's n: 0.0340

Riprap Analysis: Riprap Analysis 240 Q100- Class 3

Notes:

Input Parameters

Riprap Type: Revetment

The channel is a natural channel

Local Depth of Flow: 16.95 ft

Riprap Shape is Angular

Stability Coefficient: 0.3

This value is updated by the selected Riprap Shape

Blanket Thickness Coefficient: 1

Channel Cross-sectional Average Velocity: 4.48775 ft/s

Which is the Average Velocity with Spurs

Centerline Radius of Curvature of Channel Bend: 35 ft

Width of Water Surface at Upstream End of Channel Bend: 207.439 ft

Bank Angle: 1.5 :1 H:V

.966 < Bank Angle < 4.011

The location of the revetment protection is on the outside of a bend

Specific Gravity of Riprap: 2.65

Safety Factor: 1.1

Result Parameters

Side slope Correction Factor: 0.717953

Velocity Distribution Coefficient: 1.43756

Design Velocity: 9.61218 ft/s

Design velocity never less than average channel velocity

Computed D30: 215.557 mm

Computed D50: 258.668 mm

Riprap Class

Riprap Class Name:CLASS III

Riprap Class Order:3

The following values are an 'average' of the size fraction range for the selected riprap class.

D100: 609.60 mm

D85: 431.80 mm

D50: 317.50 mm

D15: 228.60 mm

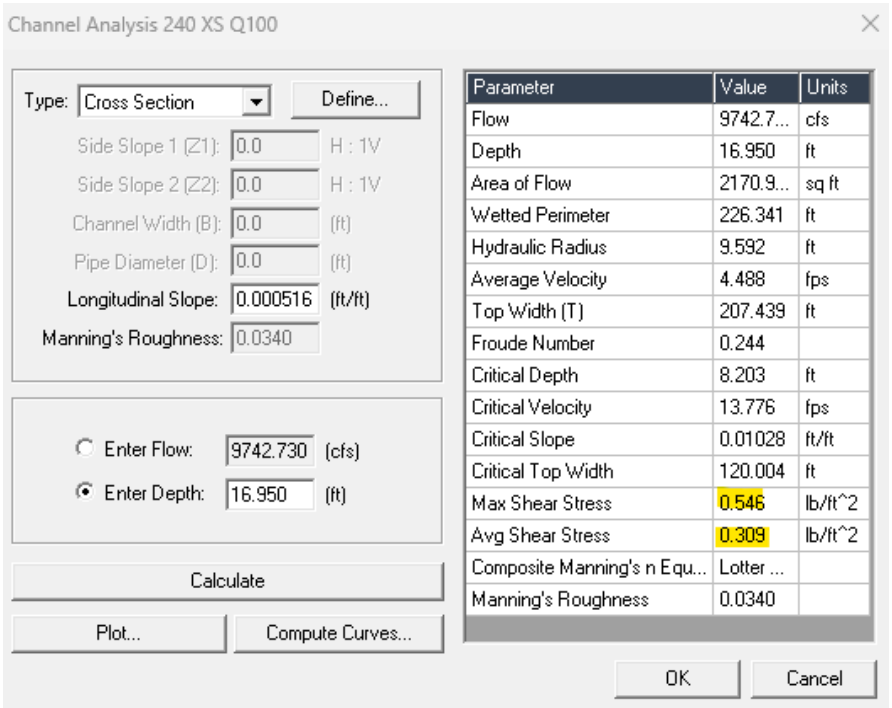
Layout Recommendations

Minimum Riprap Thickness: 288.00 in

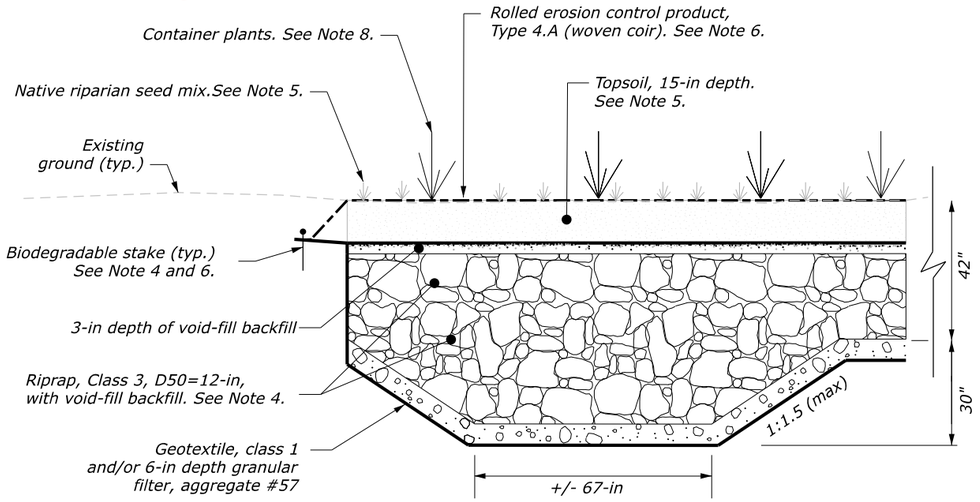
Name of Selected Channel: Channel Analysis 240 XS Q100

Riprap along banks = Class 3 – D50=12-in

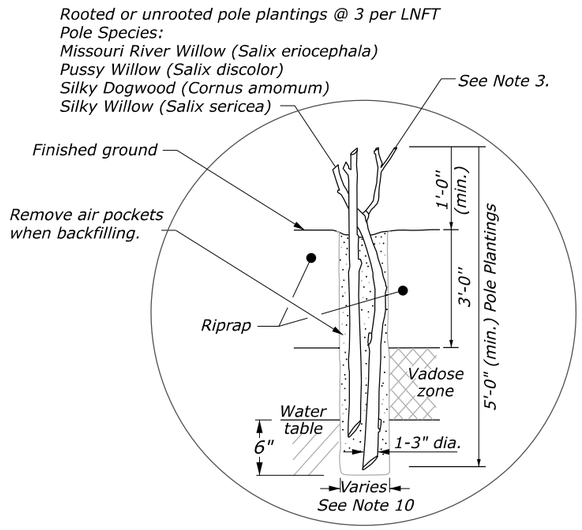
Riprap along TOE = 12”*1.5 = 18-in (Class 5)



Attachment 07:



DETAIL B
Riprap Key-in

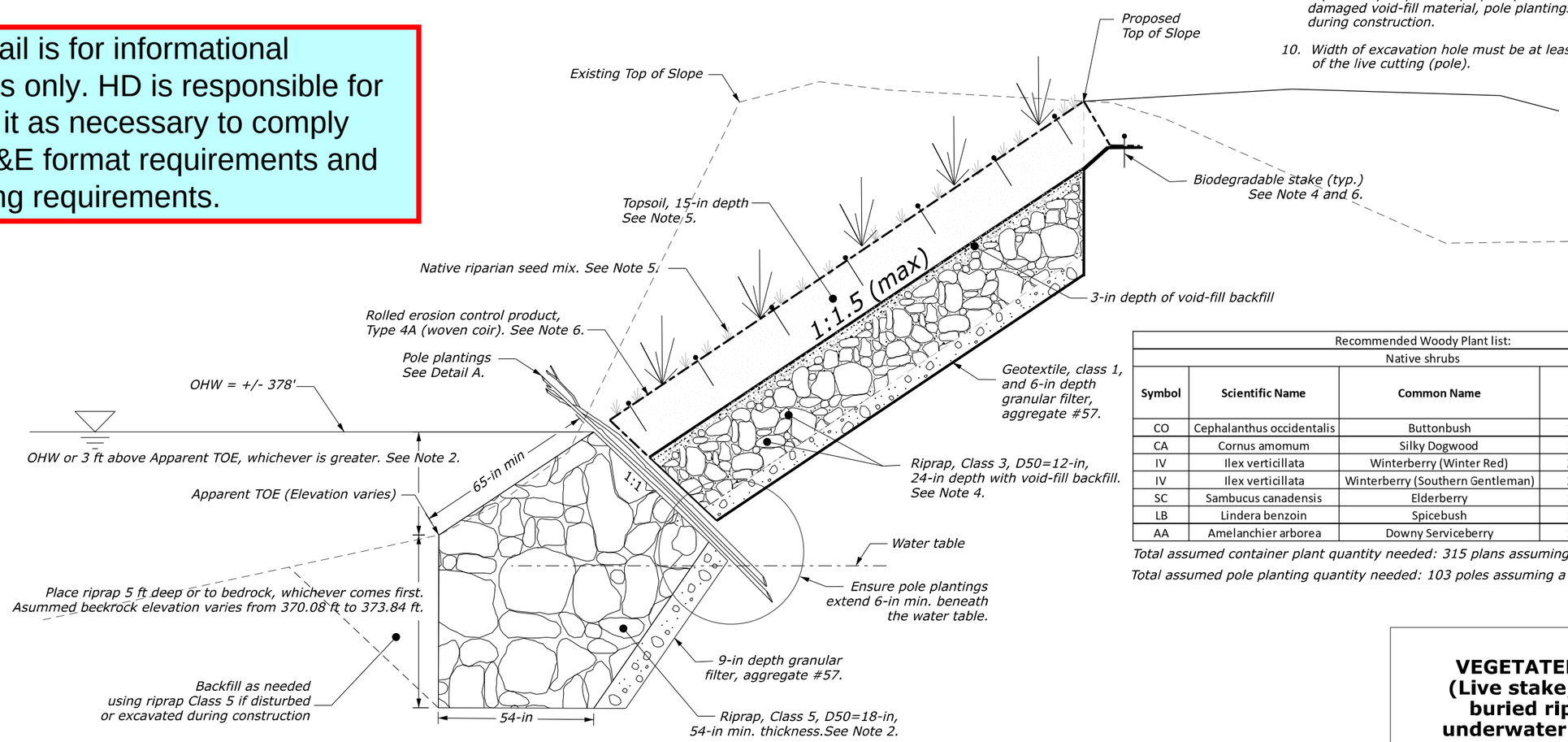


DETAIL A
POLE PLANTINGS

Notes:

1. Refer to Section 251 for more requirements.
2. Place granular filter and Class 5 riprap up to the OHWM or to a 3ft vertical rise (65-in of slope lenght), whichever is greater. Place the largest available rocks at the toe of the slope as directed by the CO.
3. Harvest and install Pole Planting on dormant season. Pole plantings harvested and not installed immediately must be either fully soaked in water, stored until they can be planted or post-planted. SSee Section 251 for more information.
4. Place geotextile, granular filter and then place riprap (class 3) while simultaneously backfilling voids with the void-fill backfill material approved by the CO. Place 3-in depth oid-fill backfill prior to encapsulate the rock with the geotextile. Key and stake the geotextile into the top of bank.
5. Place topsoil (15-in depth) and stabilize the slope using a native riparian seed mix. See Sections 624 and 625 for more information.
6. Place rolled erosion control product, type 4A (woven coir). Use bio-degradable stakes and stake the matting according to manufacturer's recommendations. Key and stake the rolled erosion control product a minimum of 12-inches into the top of bank. See Section 629 and 713 for more information.
7. Provide a key-in trench into the bank at both upstream and downstream ends to prevent flanking. See detail B.
8. Coordinate and obtain CO approval on the location/placement of all plantings prior to installation.
9. Replace any displaced riprap. Replace any displaced and damaged void-fill material, pole plantings and geotextile during construction.
10. Width of excavation hole must be at least twice the diameter of the live cutting (pole).

This detail is for informational purposes only. HD is responsible for revising it as necessary to comply with PS&E format requirements and permitting requirements.



Recommended Woody Plant list:				
Native shrubs				
Symbol	**Scientific Name**	**Common Name**	**Size**	**Recommended Spacing (In ft.)**
CO	Cephalanthus occidentalis	Buttonbush	1 Gal.	4 to 6
CA	Cornus amomum	Silky Dogwood	1 Gal.	4 to 6
IV	Ilex verticillata	Winterberry (Winter Red)	2 Gal.	3 to 5
IV	Ilex verticillata	Winterberry (Southern Gentleman)	2 Gal.	4 to 5
SC	Sambucus canadensis	Elderberry	1 Gal.	6 to 8
LB	Lindera benzoin	Spicebush	1 Gal.	4 to 6
AA	Amelanchier arborea	Downy Serviceberry	1 Gal.	6 to 8

Total assumed container plant quantity needed: 315 plans assuming a 4.5 ft spacing
Total assumed pole planting quantity needed: 103 poles assuming a 3 ft spacing

VEGETATED RIPRAP (Live stake/pole with buried riprap) for underwater conditions

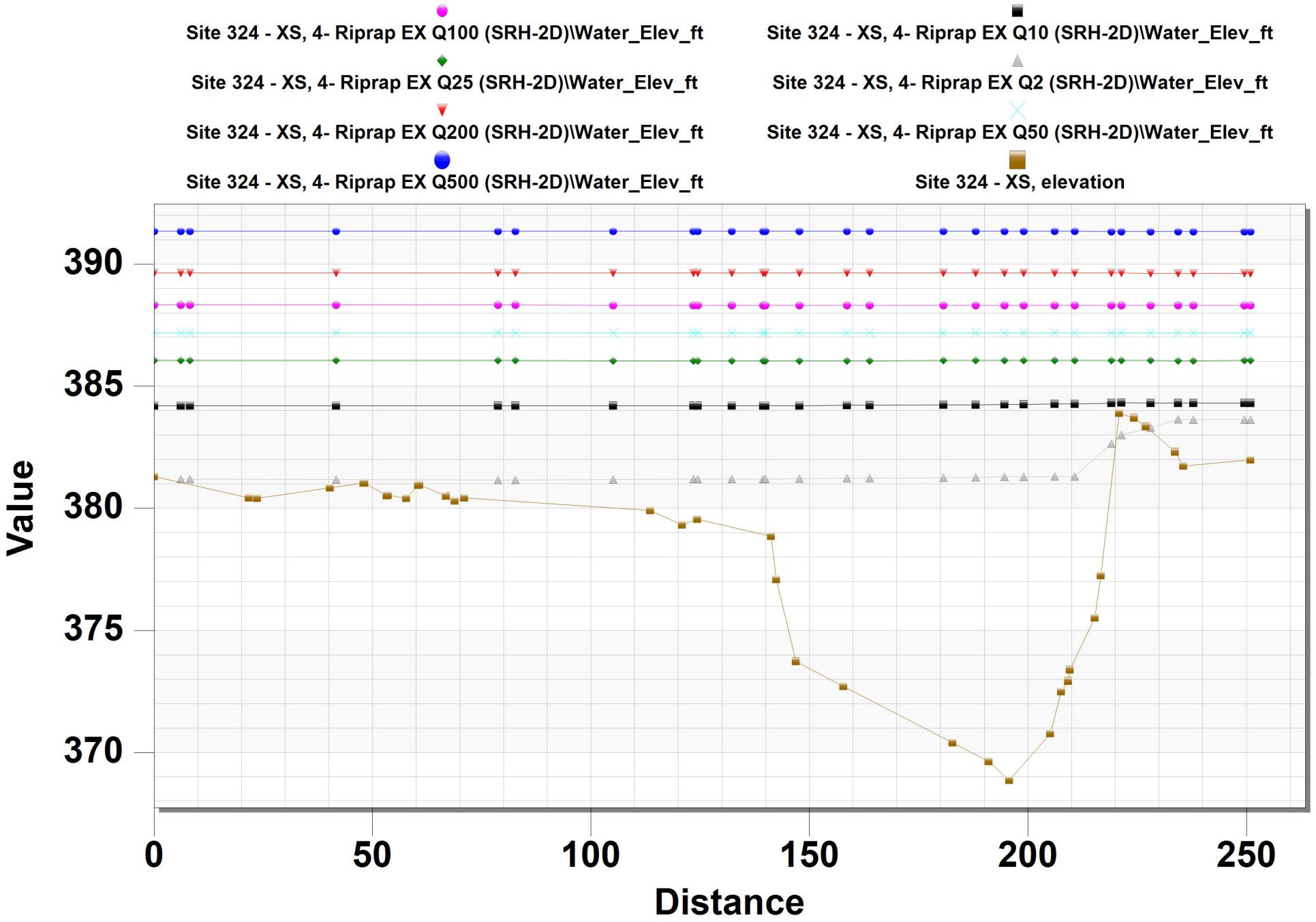
SITE 240

E251-A

Attachment 08:

NY WARI 2019-1(1)

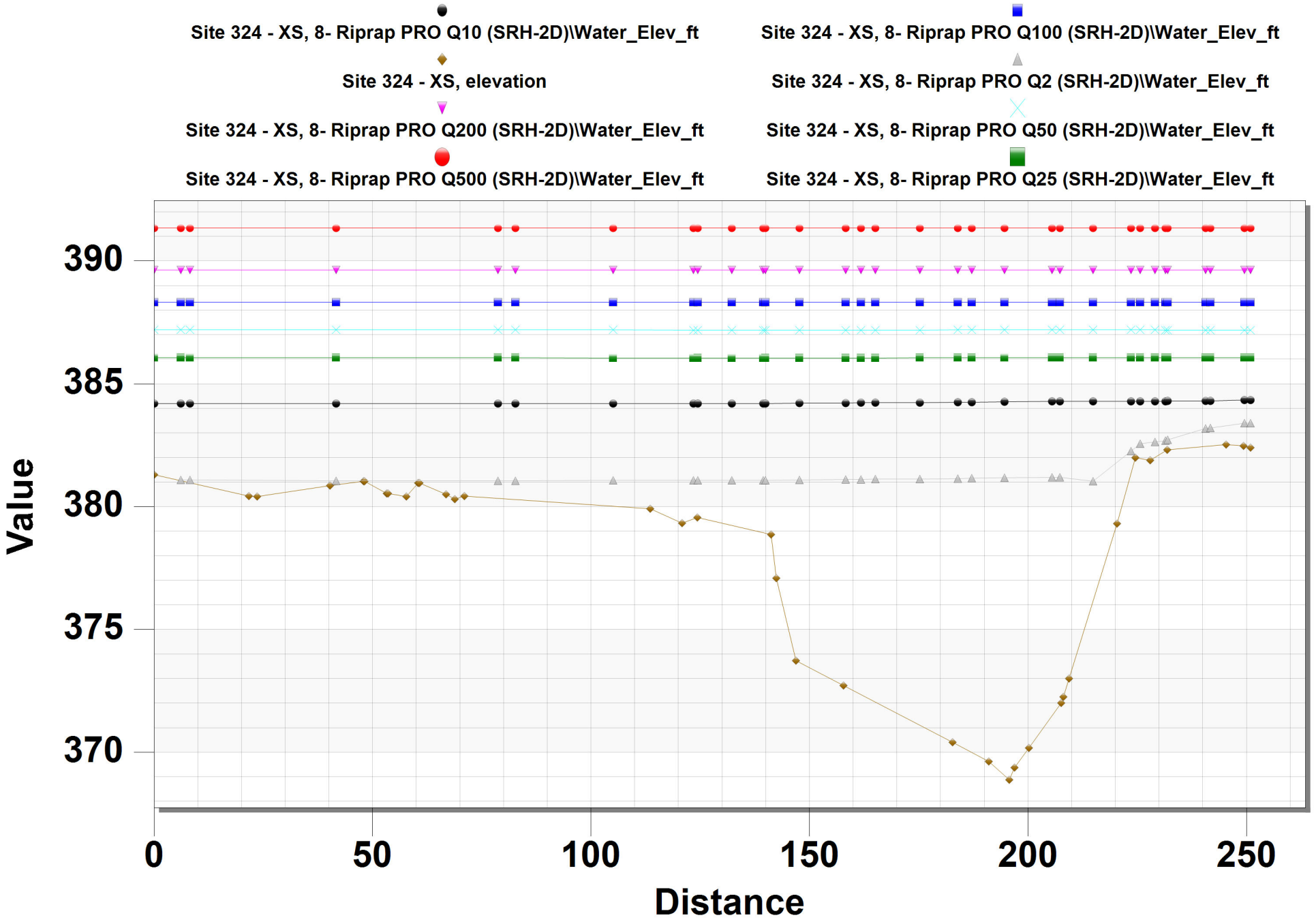
Site 324 - EX Wsels



Attachment 09:

NY WARI 2019-1(1)

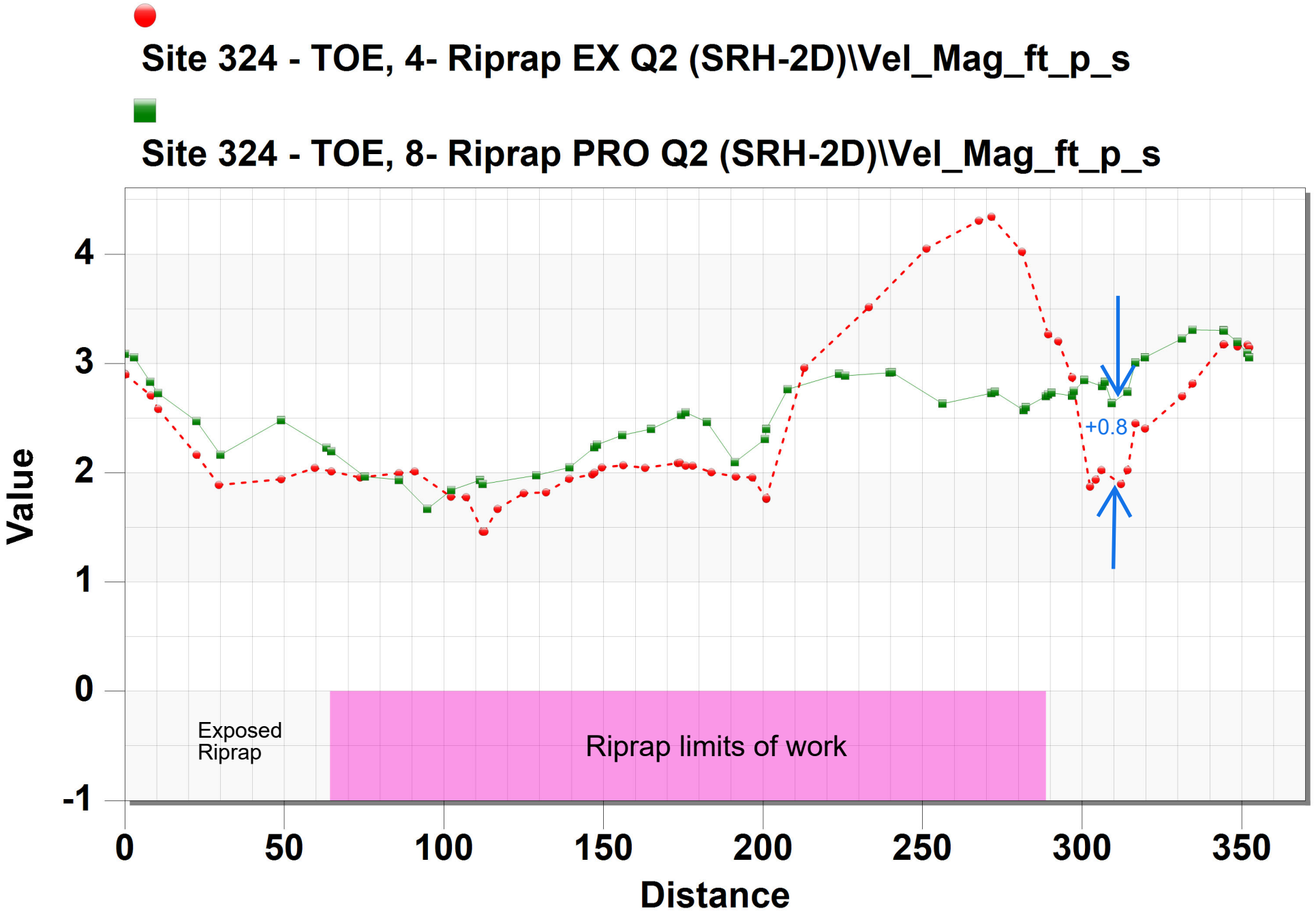
Site 324 - PRO Wsels



Attachment 10:

NY WARI 2019-1(1)

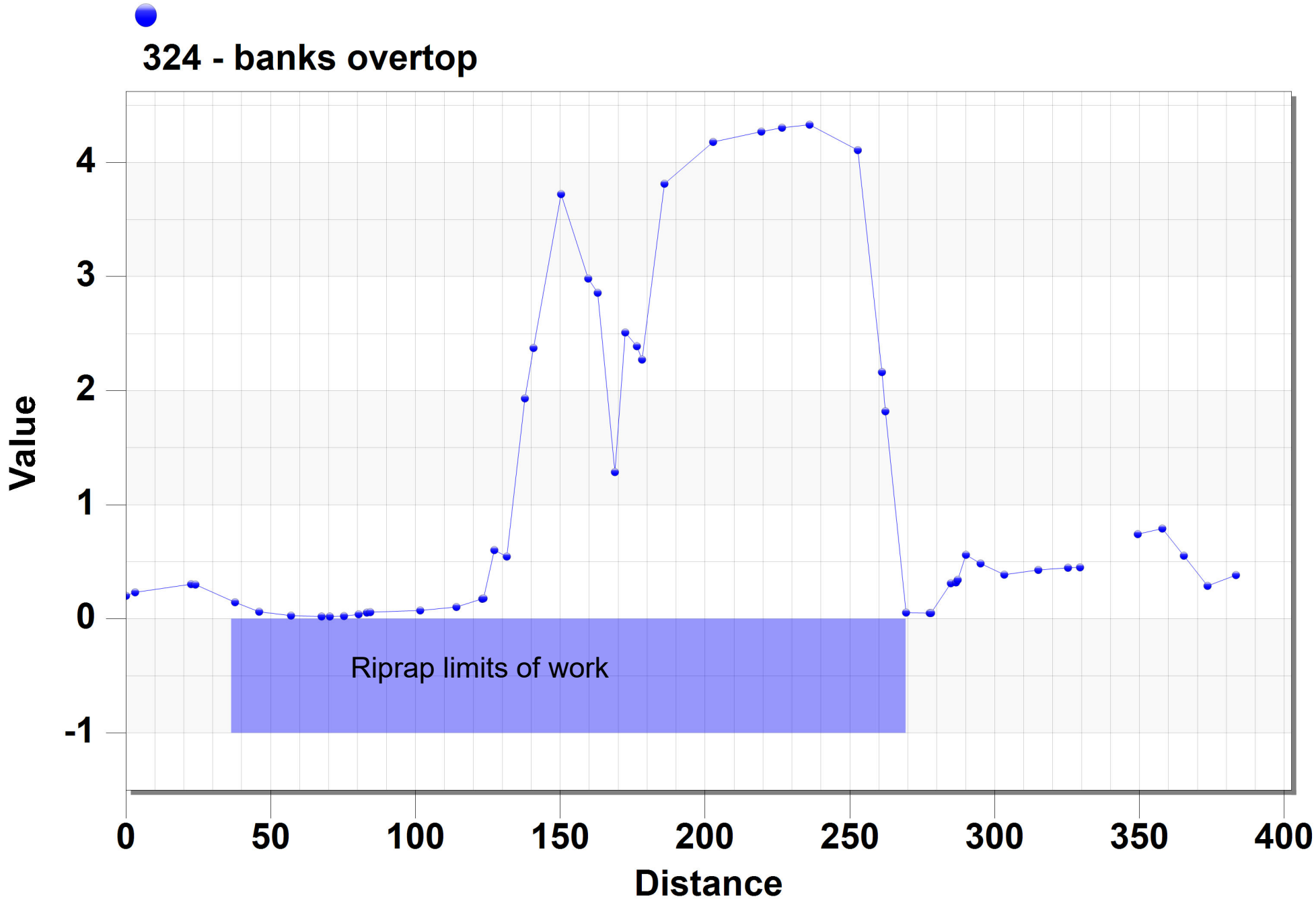
Site 324 - TOE - Velocities (fps)



Attachment 11:

NY WARI 2019-1(1)

Site 324 - Stress_lb_p_ft2 along riprap TOE



Attachment 12:

Hydraulic Analysis Report- Site 324

Project Data

Project Title: NY WARI 2019-1(1)

Designer: EFL HYD (IRV)

Project Date: Thursday, December 04, 2025

Project Units: U.S. Customary Units

Notes: Q100

Channel Analysis: Channel Analysis 324 XS Q100

Notes:

Input Parameters

Channel Type: Custom Cross Section

Cross Section Data

Station (ft	Elevation (ft)	Manning's n
0.00	380.31	0.0350
2.03	380.43	0.0350
44.94	379.92	0.0350
52.05	379.33	0.0350
67.39	378.59	0.0350
72.50	378.87	0.0350
73.69	377.11	0.0350
78.23	373.73	0.0350
89.09	372.70	0.0350
122.52	369.61	0.0350
127.01	368.87	0.0350
128.01	369.14	0.0350
128.26	369.44	0.0350
138.41	371.69	0.0350
139.80	372.50	0.0350
139.81	372.50	0.0350
141.35	373.57	0.0350
141.35	373.57	0.0350
141.36	373.57	0.0350
141.44	373.60	0.0350
141.57	373.65	0.0350
141.76	373.72	0.0350
141.90	373.77	0.0350

158.07	382.13	0.0350
160.66	383.53	0.0350
162.57	383.61	0.0350
162.67	383.61	0.0350
162.83	383.62	0.0350
164.72	383.70	0.0350
166.69	383.77	0.0350
170.24	383.63	0.0350
170.71	383.61	0.0350
170.99	383.60	0.0350
172.72	383.53	0.0350
177.30	382.40	0.0350
179.03	381.97	0.0350
181.37	381.96	0.0350
187.44	381.94	0.0350
191.90	381.93	0.0350
195.32	381.92	0.0350
195.46	381.92	0.0350
195.82	381.92	0.0350
197.05	381.91	0.0350
198.03	381.91	0.0350
246.70	381.66	0.0350
250.48	381.57	0.0350
254.66	381.59	0.0350
257.65	381.53	-----

Longitudinal Slope: 0.0005 ft/ft

Depth 21.1600 ft

Result Parameters

Flow 14406.8751 cfs

Area of Flow 2999.6622 ft^2

Wetted Perimeter 281.8309 ft

Hydraulic Radius 10.6435 ft

Average Velocity 4.8028 ft/s

Top Width 257.6550 ft

Froude Number: 0.2481

Critical Depth 13.9724 ft

Critical Velocity 12.4306 ft/s

Critical Slope: 0.0109 ft/ft

Critical Top Width 241.52 ft

Calculated Max Shear Stress 0.6602 lb/ft^2

Calculated Avg Shear Stress 0.3321 lb/ft^2

Composite Manning's n Equation: Lotter method

Manning's n: 0.0335

Riprap Analysis: Riprap Analysis 324 Q100- Class 3

Notes:

Input Parameters

Riprap Type: Revetment

The channel is a natural channel

Local Depth of Flow: 21.16 ft

Riprap Shape is Angular

Stability Coefficient: 0.3

This value is updated by the selected Riprap Shape

Blanket Thickness Coefficient: 1

Channel Cross-sectional Average Velocity: 4.80283 ft/s

Which is the Average Velocity with Spurs

Centerline Radius of Curvature of Channel Bend: 35 ft

Width of Water Surface at Upstream End of Channel Bend: 257.655 ft

Bank Angle: 1.5 :1 H:V

.966 < Bank Angle < 4.011

The location of the revetment protection is on the outside of a bend

Specific Gravity of Riprap: 2.65

Safety Factor: 1.1

Result Parameters

Side slope Correction Factor: 0.717953

Velocity Distribution Coefficient: 1.45639

Design Velocity: 10.5222 ft/s

Design velocity never less than average channel velocity

Computed D30: 259.021 mm

Computed D50: 310.825 mm

Riprap Class

Riprap Class Name:CLASS III

Riprap Class Order:3

The following values are an 'average' of the size fraction range for the selected riprap class.

D100: 609.60 mm

D85: 431.80 mm

D50: 317.50 mm

D15: 228.60 mm

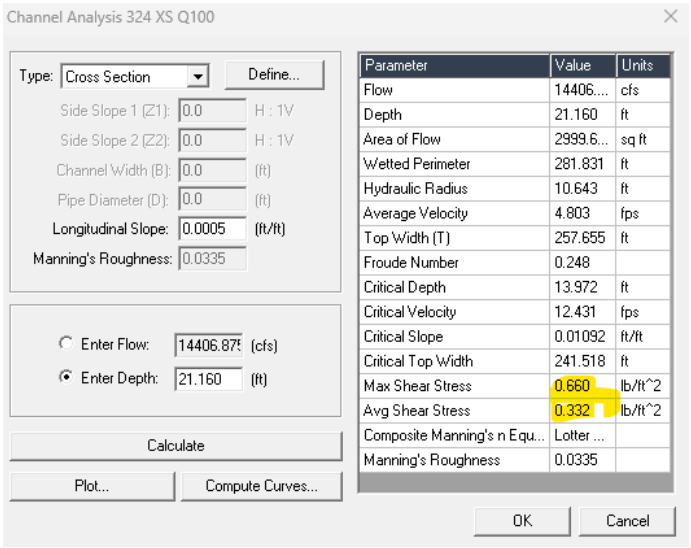
Layout Recommendations

Minimum Riprap Thickness: 288.00 in

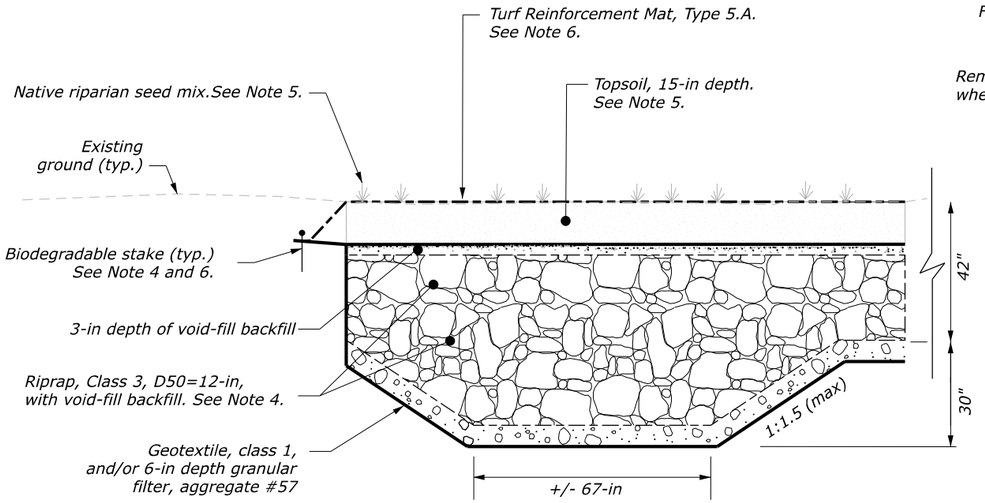
Name of Selected Channel: Channel Analysis 324 XS Q100

Riprap along banks = Class 3 – D50=12-in

Riprap along TOE = 12”*1.5 = 18-in (Class 5)

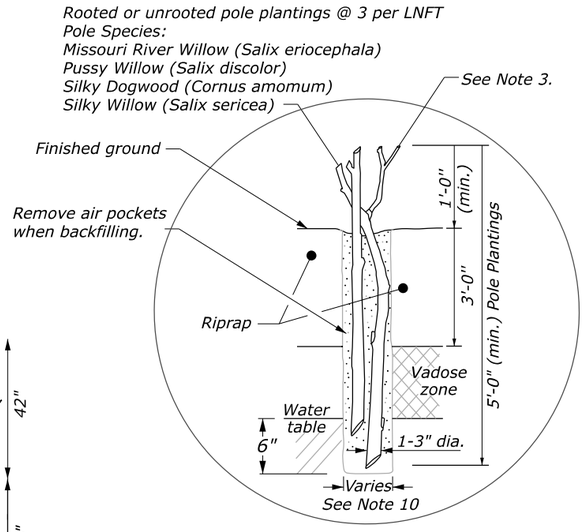


Attachment 13a:



DETAIL B

Riprap Key-in



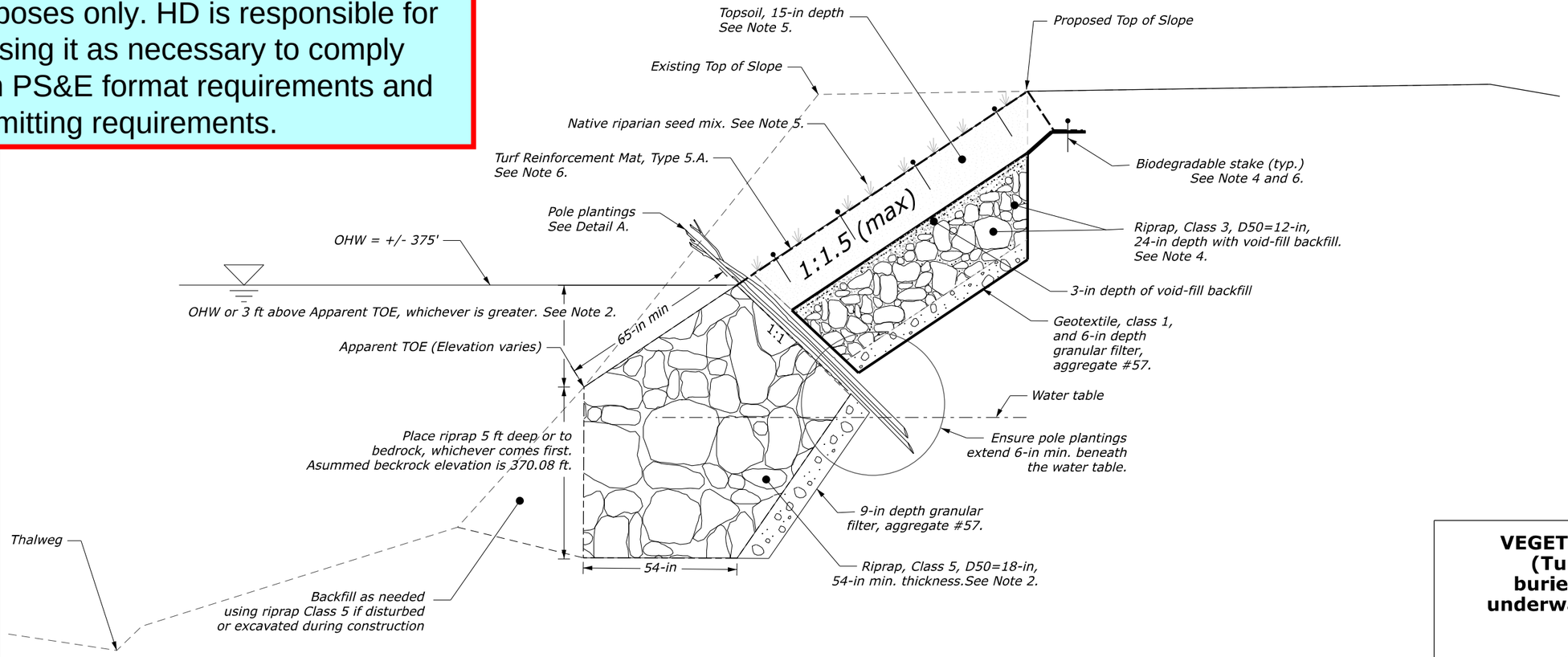
DETAIL A

POLE PLANTINGS

Notes:

1. Refer to Section 251 for more requirements.
2. Place granular filter and Class 5 riprap up to the OHWM or to a 3ft vertical rise (65-in of slope lenght), whichever is greater. Place the largest available rocks at the toe of the slope as directed by the CO.
3. Harvest and install Pole Planting on dormant season. Pole plantings harvested and not installed immediately must be either fully soaked in water, stored until they can be planted or post-planted. SSee Section 251 for more information.
4. Place geotextile, granular filter and then place riprap (class 3) while simultaneously backfilling voids with the void-fill backfill material approved by the CO. Place 3-in depth oid-fill backfill prior to encapsulate the rock with the geotextile. Key and stake the geotextile into the top of bank.
5. Place topsoil (15-in depth) and stabilize the slope using a native riparian seed mix. See Sections 624 and 625 for more information.
6. Place Turf Reinforcement Mat, type 5A. Use bio-degradable stakes and stake the matting according to manufacturer's recommendations. Key and stake the turf reinforcement mat product a minimum of 12-inches into the top of bank. See Section 629 and 713 for more information.
7. Provide a key-in trench into the bank at both upstream and downstream ends to prevent flanking. See detail B.
8. Coordinate and obtain CO approval on the location/placement of all plantings prior to installation.
9. Replace any displaced riprap. Replace any displaced and damaged void-fill material, pole plantings and geotextile during construction.
10. Width of excavation hole must be at least twice the diameter of the live cutting (pole).

This detail is for informational purposes only. HD is responsible for revising it as necessary to comply with PS&E format requirements and permitting requirements.

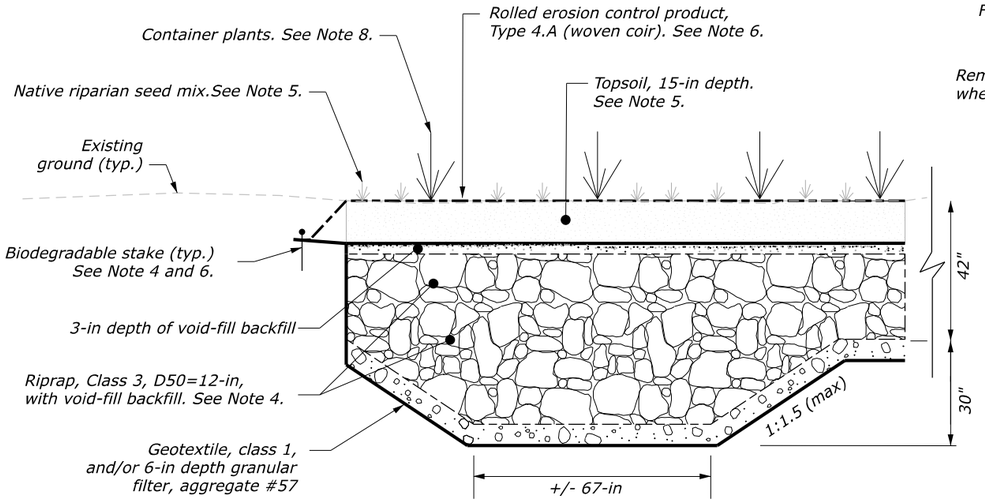


VEGETATED RIPRAP (Turf Mat with buried riprap) for underwater conditions ALT 1

SITE 324

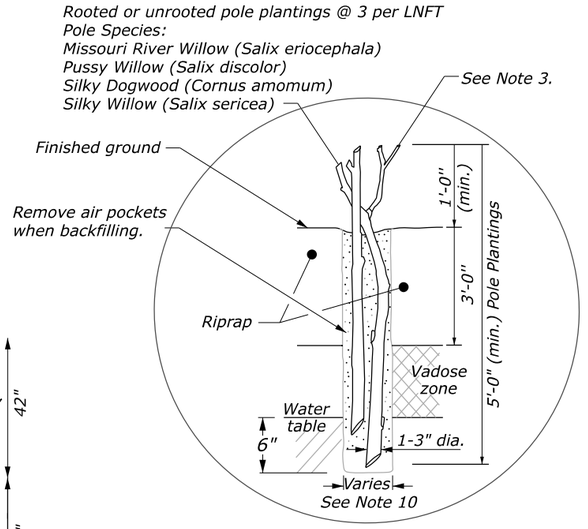
E251-B

Attachment 13b:



DETAIL B

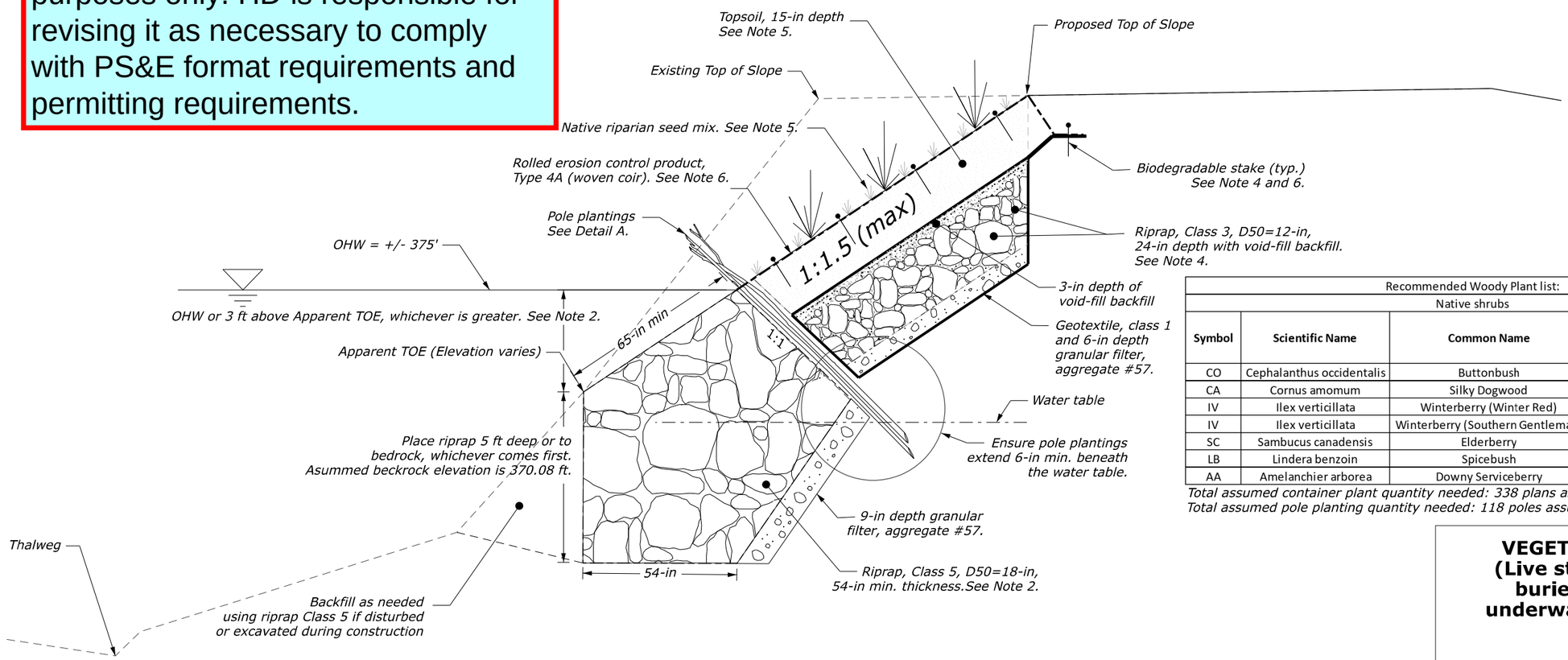
Riprap Key-in



DETAIL A

POLE PLANTINGS

This detail is for informational purposes only. HD is responsible for revising it as necessary to comply with PS&E format requirements and permitting requirements.



Notes:

1. Refer to Section 251 for more requirements.
2. Place granular filter and Class 5 riprap up to the OHWM or to a 3ft vertical rise (65-in of slope lenght), whichever is greater. Place the largest available rocks at the toe of the slope as directed by the CO.
3. Harvest and install Pole Planting on dormant season. Pole plantings harvested and not installed immediately must be either fully soaked in water, stored until they can be planted or post-planted. SSee Section 251 for more information.
4. Place geotextile, granular filter and then place riprap (class 3) while simultaneously backfilling voids with the void-fill backfill material approved by the CO. Place 3-in depth oid-fill backfill prior to encapsulate the rock with the geotextile. Key and stake the geotextile into the top of bank.
5. Place topsoil (15-in depth) and stabilize the slope using a native riparian seed mix. See Sections 624 and 625 for more information.
6. Place rolled erosion control product, type 4A (woven coir). Use bio-degradable stakes and stake the matting according to manufacturer's recommendations. Key and stake the rolled erosion control product a minimum of 12-inches into the top of bank. See Section 629 and 713 for more information.
7. Provide a key-in trench into the bank at both upstream and downstream ends to prevent flanking. See detail B.
8. Coordinate and obtain CO approval on the location/placement of all plantings prior to installation.
9. Replace any displaced riprap. Replace any displaced and damaged void-fill material, pole plantings and geotextile during construction.
10. Width of excavation hole must be at least twice the diameter of the live cutting (pole).

Recommended Woody Plant list:				
Native shrubs				
Symbol	**Scientific Name**	**Common Name**	**Size**	**Recommended Spacing (In ft.)**
CO	Cephalanthus occidentalis	Buttonbush	1 Gal.	4 to 6
CA	Cornus amomum	Silky Dogwood	1 Gal.	4 to 6
IV	Ilex verticillata	Winterberry (Winter Red)	2 Gal.	3 to 5
IV	Ilex verticillata	Winterberry (Southern Gentleman)	2 Gal.	4 to 5
SC	Sambucus canadensis	Elderberry	1 Gal.	6 to 8
LB	Lindera benzoin	Spicebush	1 Gal.	4 to 6
AA	Amelanchier arborea	Downy Serviceberry	1 Gal.	6 to 8

Total assumed container plant quantity needed: 338 plans assuming a 4.5 ft spacing
Total assumed pole planting quantity needed: 118 poles assuming a 3 ft spacing

VEGETATED RIPRAP (Live stake/pole with buried riprap) for underwater conditions. ALT 2

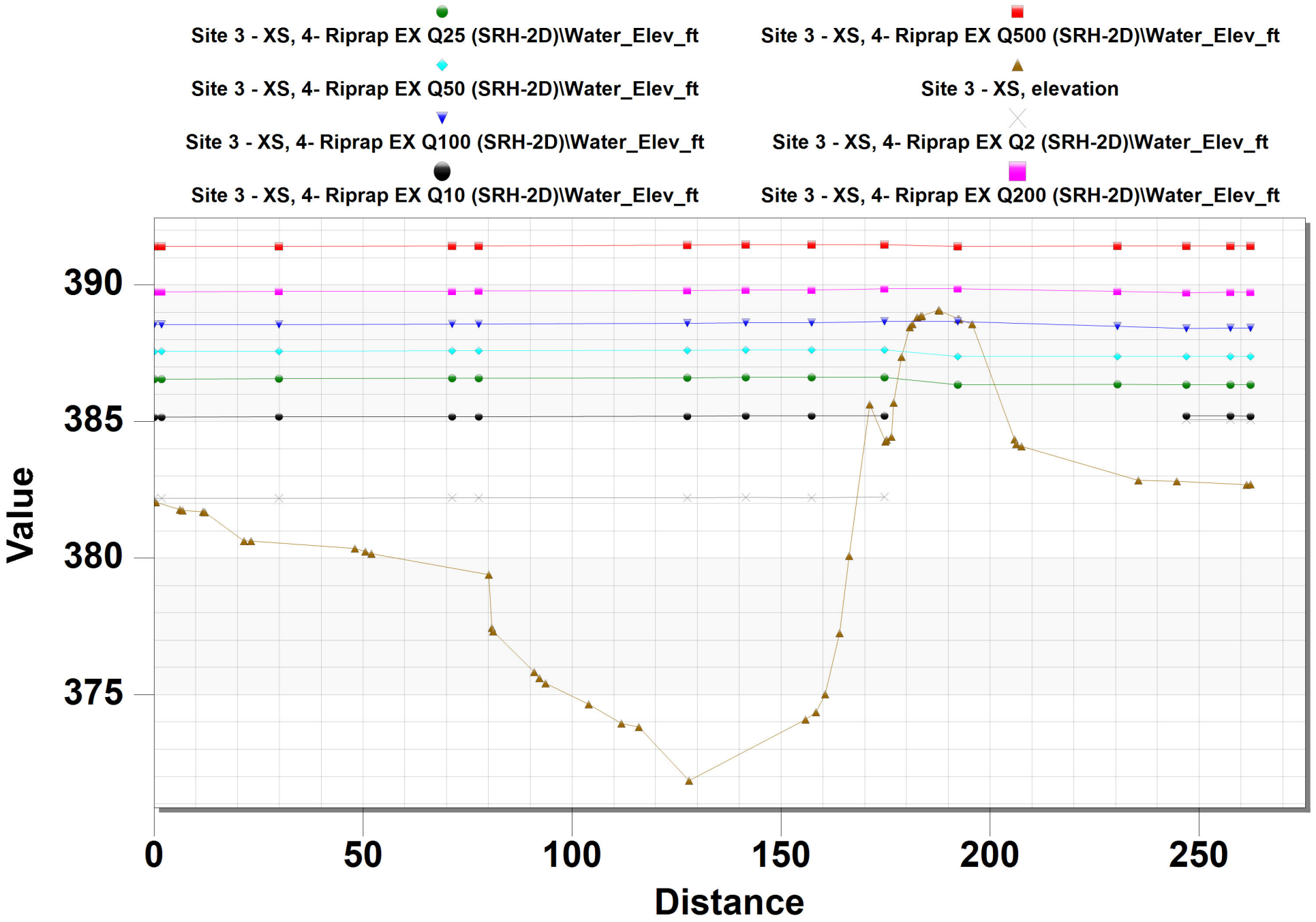
SITE 324

E251-C

Attachment 14:

NY WARI 2019-1(1)

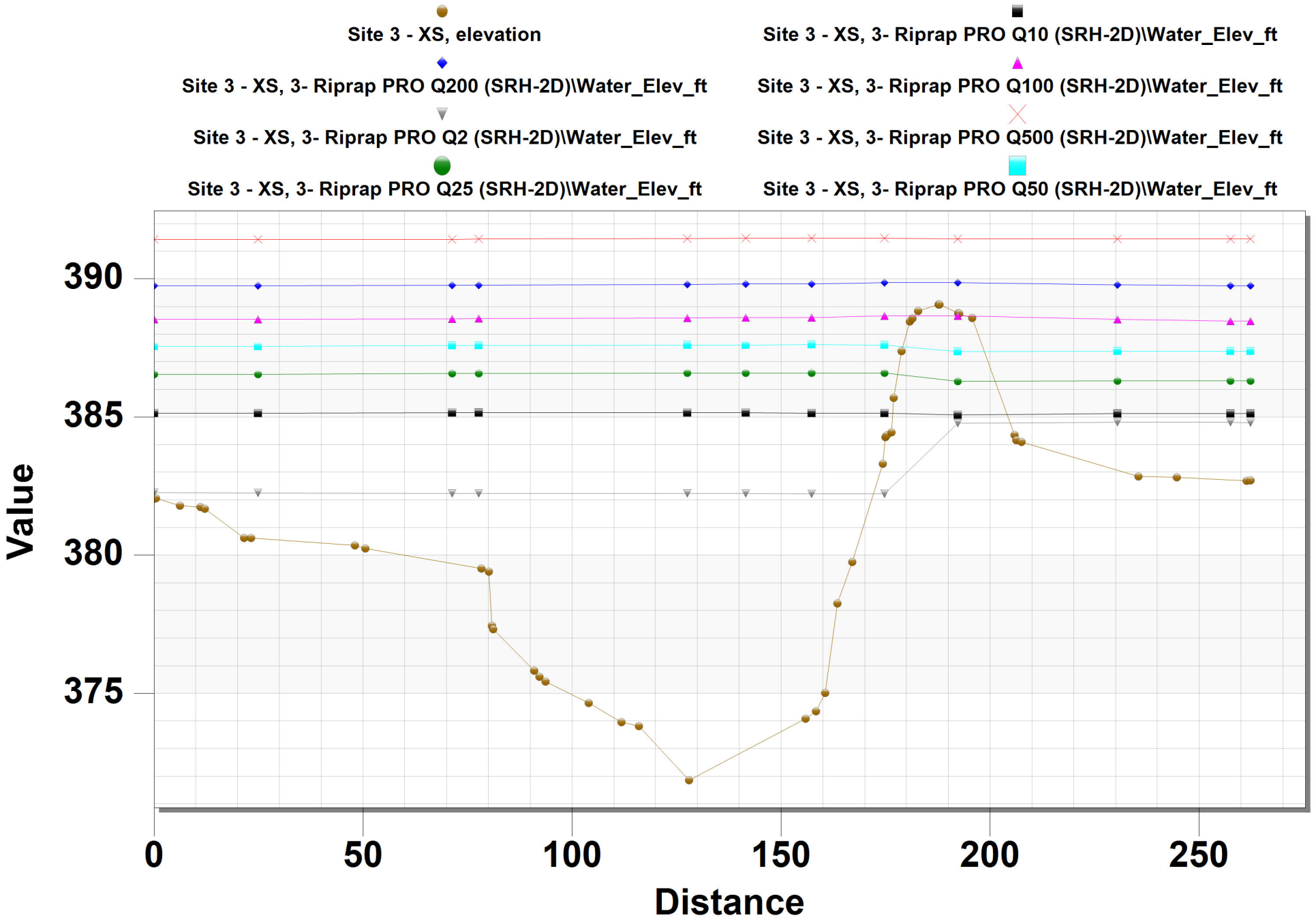
Site 3 XS - EX Wsels



Attachment 15:

NY WARI 2019-1(1)

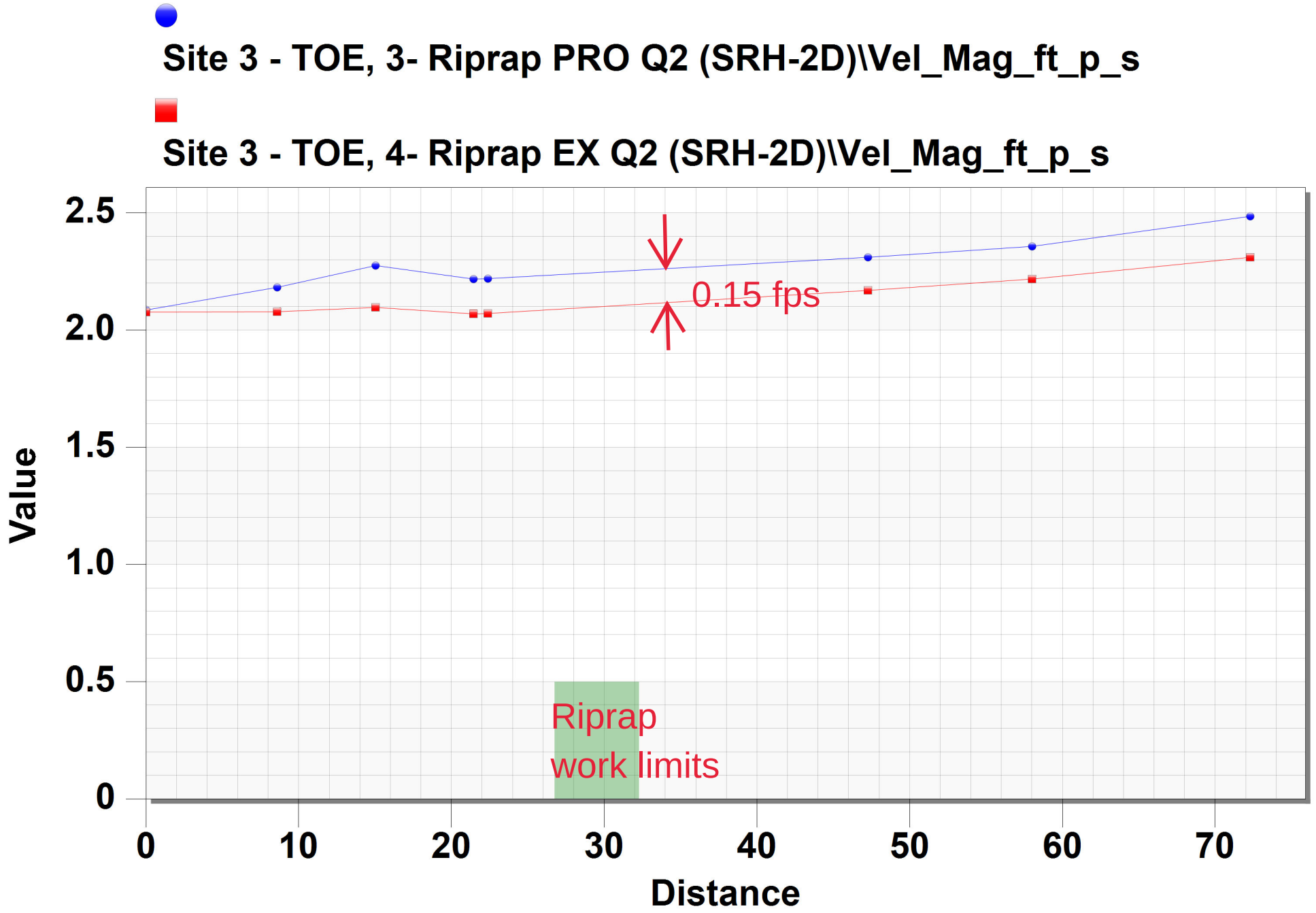
Site 3 - PRO Wsels



Attachment 16:

NY WARI 2019-1(1)

Site 3 - Vel_Mag_ft_p_s



Attachment 17

Site 3 - Hydraulic Analysis Report

Project Data

Project Title: NY WARI 2019-1(1)

Designer: EFL

Project Date: Friday, May 13, 2022

Project Units: U.S. Customary Units

Notes:

Channel Analysis: Channel Analysis Q100

Notes:

Input Parameters

Channel Type: Custom Cross Section

Cross Section Data

Station (ft	Elevation (ft)	Manning's n
0.00	382.05	0.0400
0.43	382.06	0.0400
6.19	381.78	0.0400
6.82	381.75	0.0400
11.73	381.70	0.0400
12.17	381.68	0.0400
21.55	380.62	0.0400
23.21	380.62	0.0400
48.06	380.36	0.0400
50.60	380.25	0.0400
52.01	380.16	0.0400
80.14	379.40	0.0400
80.87	377.44	0.0400
81.11	377.31	0.0400
91.00	375.82	0.0400
92.23	375.60	0.0400
93.62	375.42	0.0400
103.94	374.65	0.0400
111.82	373.95	0.0400
116.00	373.82	0.0400
127.94	371.85	0.0400
155.90	374.09	0.0400
158.41	374.36	0.0400
160.55	375.01	0.0400

163.98	377.26	0.0400
166.28	380.09	0.0400
171.16	385.63	0.0400
174.90	384.27	0.0400
175.28	384.33	0.0400
176.43	384.45	0.0400
176.97	385.69	0.0400
178.82	387.38	0.0400
180.80	388.45	0.0400
181.39	388.57	0.0400
182.59	388.83	0.0400
183.38	388.85	0.0400
183.72	388.87	0.0400
187.71	389.08	0.0400
187.92	389.07	0.0400
192.37	388.75	0.0400
192.60	388.74	0.0400
195.75	388.58	0.0400
205.81	384.34	0.0400
206.25	384.17	0.0400
207.49	384.10	0.0400
235.43	382.85	0.0400
244.63	382.82	0.0400
261.34	382.68	0.0400
262.28	382.70	-----

Longitudinal Slope: 0.0013 ft/ft

Flow 7470.0000 cfs

Result Parameters

Depth 13.9946 ft

Area of Flow 1598.3118 ft^2

Wetted Perimeter 252.1603 ft

Hydraulic Radius 6.3385 ft

Average Velocity 4.6737 ft/s

Top Width 237.1655 ft

Froude Number: 0.3173

Critical Depth 9.1936 ft

Critical Velocity 11.7225 ft/s

Critical Slope: 0.0150 ft/ft

Critical Top Width 149.32 ft

Calculated Max Shear Stress 1.1352 lb/ft^2

Calculated Avg Shear Stress 0.5142 lb/ft^2

Composite Manning's n Equation: Lotter method

Manning's n: 0.0393

Riprap Analysis: Riprap Analysis Q100

Notes:

Input Parameters

Riprap Type: Revetment

The channel is a natural channel

Local Depth of Flow: 13.9946 ft

Riprap Shape is Angular

Stability Coefficient: 0.3

This value is updated by the selected Riprap Shape

Blanket Thickness Coefficient: 1

Channel Cross-sectional Average Velocity: 4.67368 ft/s

Which is the Average Velocity with Spurs

Centerline Radius of Curvature of Channel Bend: 1e+09 ft

Width of Water Surface at Upstream End of Channel Bend: 237.165 ft

Bank Angle: 1.5 :1 H:V

.966 < Bank Angle < 4.011

The location of the revetment protection is on a straight channel

Specific Gravity of Riprap: 2.65

Safety Factor: 1.1

Result Parameters

Side slope Correction Factor: 0.717953

Velocity Distribution Coefficient: 1

Design Velocity: 4.67368 ft/s

Design velocity never less than average channel velocity

Computed D30: 25.9316 mm

Computed D50: 31.118 mm

Riprap Class

Riprap Class Name:CLASS I

Riprap Class Order:1

The following values are an 'average' of the size fraction range for the selected riprap class.

D100: 304.80 mm

D85: 228.60 mm

D50: 165.10 mm

D15: 114.30 mm

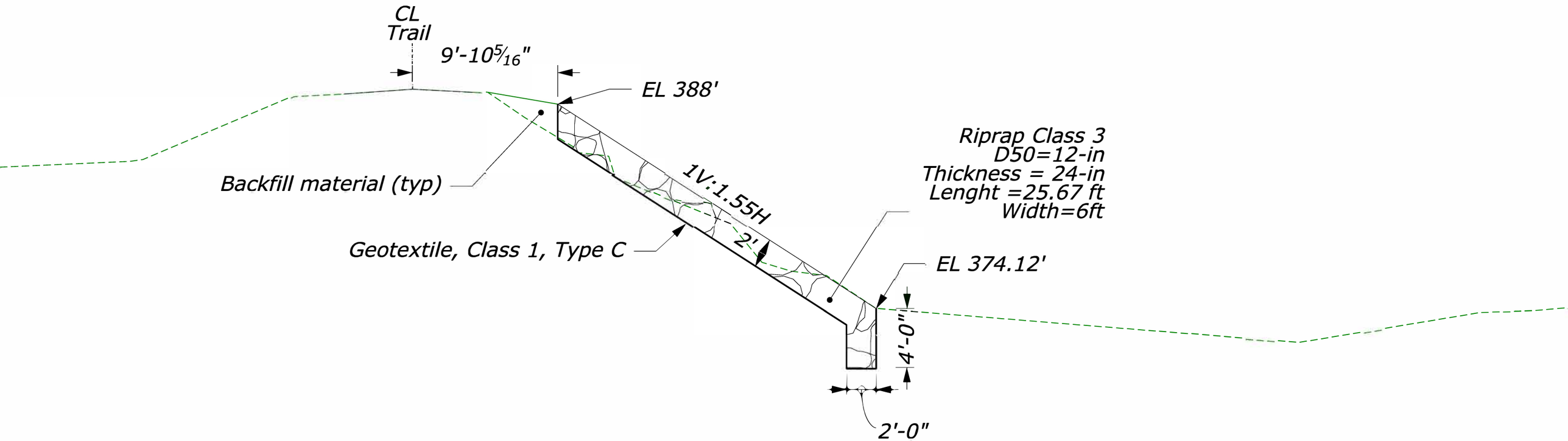
Layout Recommendations

Minimum Riprap Thickness: 144.00 in

EFL Recommendation:

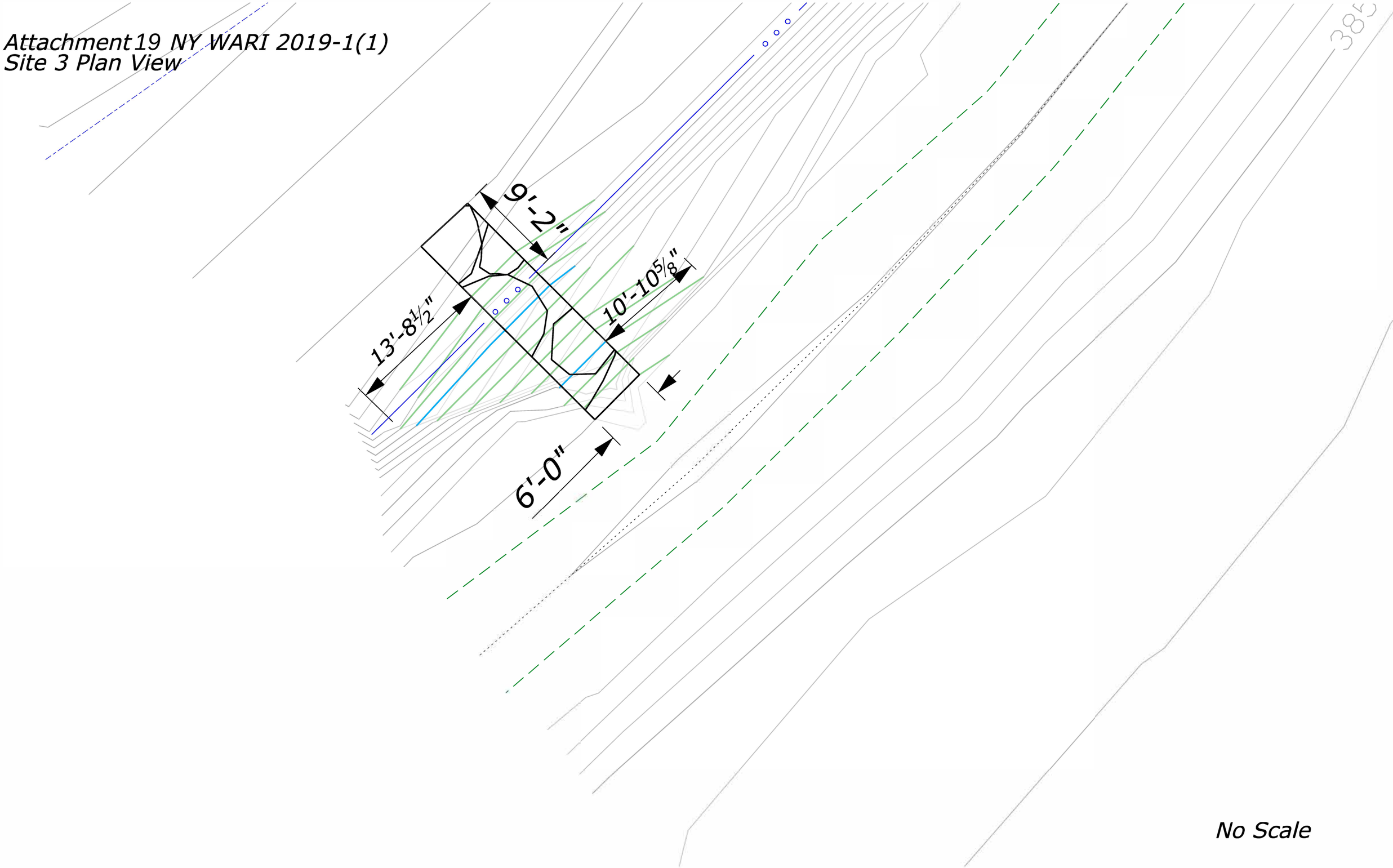
Based on our analysis, the minimum recommended riprap size is Class 1. However, this size may be too small for adequate protection. To ensure greater stability and performance, we recommend using Riprap Class 3 at Site 3.

Attachment #18: NY WARI 2019-1(1) Site 3 (Gully repair) Cross Section



No Scale

Attachment 19 NY WARI 2019-1(1)
Site 3 Plan View



No Scale

NOTES TO USERS

This map is for use in administering the National Flood Insurance Program. It does not necessarily identify all areas subject to flooding, particularly from local drainage sources of small size. The **community map repository** should be consulted for possible updated or additional flood hazard information.

To obtain more detailed information in areas where **Base Flood Elevations** (BFEs) and/or **floodways** have been determined, users are encouraged to consult the Flood Profiles and Floodway Data and/or Summary of Stillwater Elevations tables contained within the Flood Insurance Study (FIS) report that accompanies this FIRM. Users should be aware that BFEs shown on the FIRM represent rounded whole-foot elevations. These BFEs are intended for flood insurance rating purposes only and should not be used as the sole source of flood elevation information. Accordingly, flood elevation data presented in the FIS report should be utilized in conjunction with the FIRM for purposes of construction and/or floodplain management.

Coastal Base Flood Elevations shown on this map apply only landward of 0.0' North American Vertical Datum of 1988 (NAVD 88). Users of this FIRM should be aware that coastal flood elevations are also provided in the Summary of Stillwater Elevations tables in the Flood Insurance Study report for this jurisdiction. Elevations shown in the Summary of Stillwater Elevations tables should be used for construction and/or floodplain management purposes when they are higher than the elevations shown on this FIRM.

Boundaries of the **floodways** were computed at cross sections and interpolated between cross sections. The floodways were based on hydraulic considerations with regard to requirements of the National Flood Insurance Program. Floodway widths and other pertinent floodway data are provided in the Flood Insurance Study report for this jurisdiction.

Certain areas not in Special Flood Hazard Areas may be protected by **flood control structures**. Refer to Section 2.4 "Flood Protection Measures" of the Flood Insurance Study report for information on flood control structures for this jurisdiction.

The **projection** used in the preparation of this map was Universal Transverse Mercator (UTM) zone 18. The **horizontal datum** was NAD 83, GRS80 spheroid. Differences in datum, spheroid, projection or UTM zones used in the production of FIRMs for adjacent jurisdictions may result in slight positional differences in map features across jurisdiction boundaries. These differences do not affect the accuracy of this FIRM.

Flood elevations on this map are referenced to the North American Vertical Datum of 1988. These flood elevations must be compared to structure and ground elevations referenced to the same **vertical datum**. For information regarding conversion between the National Geodetic Vertical Datum of 1929 and the North American Vertical Datum of 1988, visit the National Geodetic Survey website at http://www.ngs.noaa.gov or contact the National Geodetic Survey at the following address:

NGS Information Services
NOAA, N/NGS12
National Geodetic Survey
SSMC-3, #9202
1315 East-West Highway
Silver Spring, Maryland 20910-3282
(301) 713-3242

To obtain current elevation, description, and/or location information for **bench marks** shown on this map, please contact the Information Services Branch of the National Geodetic Survey at **(301) 713-3242**, or visit its website at http://www.ngs.noaa.gov.

Base map information shown on this FIRM was derived from digital orthophotography provided by the New York Sate Office of Cyber Security & Critical Infrastructure Coordination. This information was provided as 30-centimeter and 60-centimeter resolution natural color orthoimagery from photography dated April-May 2004.

Based on updated topographic information, this map reflects more detailed and up-to-date **stream channel configurations and floodplain delineations** than those shown on the previous FIRM for this jurisdiction. As a result, the Flood Profiles and Floodway Data tables in the Flood Insurance Study Report (which contains authoritative hydraulic data) may reflect stream channel distances that differ from what is shown on this map. Also, the road to floodplain relationships for unrevised streams may differ from what is shown on previous maps.

Corporate limits shown on this map are based on the best data available at the time of publication. Because changes due to annexations or de-annexations may have occurred after this map was published, map users should contact appropriate community officials to verify current corporate limit locations.

Please refer to the separately printed **Map Index** for an overview map of the county showing the layout of map panels; community map repository addresses; and a Listing of Communities table containing National Flood Insurance Program dates for each community as well as a listing of the panels on which each community is located.

Contact the **FEMA Map Service Center** at 1-800-358-9616 for information on available products associated with this FIRM. Available products may include previously issued Letters of Map Change, a Flood Insurance Study report, and/or digital versions of this map. The FEMA Map Service Center may also be reached by Fax at 1-800-358-9620 and its website at http://msc.fema.gov.

If you have **questions about this map** or questions concerning the National Flood Insurance Program in general, please call **1-877-FEMA MAP** (1-877-336-2627) or visit the FEMA website at http://www.fema.gov.



LEGEND

SPECIAL FLOOD HAZARD AREAS SUBJECT TO INUNDATION BY THE 1% ANNUAL CHANCE FLOOD

The 1% annual flood (100-year flood), also known as the base flood, is the flood that has a 1% chance of being equaled or exceeded in any given year. The Special Flood Hazard Area is the area subject to flooding by the 1% annual chance flood. Areas of Special Flood Hazard include Zones A, AE, AH, AO, AR, A99, V, and VE. The Base Flood Elevation is the water-surface elevation of the 1% annual chance flood.

ZONE A	No Base Flood Elevations determined.
ZONE AE	Base Flood Elevations determined.
ZONE AH	Flood depths of 1 to 3 feet (usually areas of ponding); Base Flood Elevations determined.
ZONE AO	Flood depths of 1 to 3 feet (usually sheet flow on sloping terrain); average depths determined. For areas of alluvial fan flooding, velocities also determined.
ZONE AR	Special Flood Hazard Area formerly protected from the 1% annual chance flood by a flood control system that was subsequently decertified. Zone AR indicates that the former flood control system is being restored to provide protection from the 1% annual chance or greater flood.
ZONE A99	Area to be protected from 1% annual chance flood by a Federal flood protection system under construction; no Base Flood Elevations determined.
ZONE V	Coastal flood zone with velocity hazard (wave action); no Base Flood Elevations determined.
ZONE VE	Coastal flood zone with velocity hazard (wave action); Base Flood Elevations determined.

FLOODWAY AREAS IN ZONE AE

The floodway is the channel of a stream plus any adjacent floodplain areas that must be kept free of encroachment so that the 1% annual chance flood can be carried without substantial increases in flood heights.

OTHER FLOOD AREAS

ZONE X	Areas of 0.2% annual chance flood; areas of 1% annual chance flood with average depths of less than 1 foot or with drainage areas less than 1 square mile; and areas protected by levees from 1% annual chance flood.

OTHER AREAS

ZONE X	Areas determined to be outside the 0.2% annual chance floodplain.
ZONE D	Areas in which flood hazards are undetermined, but possible.

COASTAL BARRIER RESOURCES SYSTEM (CBRS) AREAS

OTHERWISE PROTECTED AREAS (OPAs)

CBRS areas and OPAs are normally located within or adjacent to Special Flood Hazard Areas.

- 1% annual chance floodplain boundary
- 0.2% annual chance floodplain boundary
- Floodway boundary
- Zone D boundary
- CBRS and OPA boundary
- Boundary dividing Special Flood Hazard Area Zones and boundary dividing Special Flood Hazard Areas of different Base Flood Elevations, flood depths or flood velocities.
- 513 — Base Flood Elevation line and value; elevation in feet*
- (EL 987) — Base Flood Elevation value where uniform within zone; elevation in feet*

* Referenced to the North American Vertical Datum of 1988

- A — A — Cross section line
- 23 — 23 — Limited detail cross section line
- 97 — 97 — Transect line
- 87°07'45", 32°22'30" — Geographic coordinates referenced to the North American Datum of 1983 (NAD 83), Western Hemisphere
- 4276000mN — 1000-meter Universal Transverse Mercator grid values, zone 18N
- 600000 FT — 5000-foot grid ticks: New York State Plane coordinate system, East zone (FIPSZONE 3101), Transverse Mercator projection
- DX5510 × — Bench mark (see explanation in Notes to Users section of this FIRM panel)
- M1.5 — River Mile

MAP REPOSITORY
Refer to listing of Map Repositories on Map Index

EFFECTIVE DATE OF COUNTYWIDE FLOOD INSURANCE RATE MAP
August 3, 2009

EFFECTIVE DATE(S) OF REVISION(S) TO THIS PANEL

For community map revision history prior to countywide mapping, refer to the Community Map History table located in the Flood Insurance Study report for this jurisdiction.

To determine if flood insurance is available in this community, contact your Insurance agent or call the National Flood Insurance Program at 1-800-638-6620.

MAP SCALE 1" = 500'
250 0 500 1000 FEET
100 0 150 300 METERS

NFIP
NATIONAL FLOOD INSURANCE PROGRAM

PANEL 0417E

FIRM

FLOOD INSURANCE RATE MAP

for ORANGE COUNTY, NEW YORK
(ALL JURISDICTIONS)

CONTAINS:

COMMUNITY	NUMBER
MINISINK, TOWN OF	360620
WARWICK, TOWN OF	360636

PANEL 417 OF 630
MAP SUFFIX: E
(SEE MAP INDEX FOR FIRM PANEL LAYOUT)

Notice to User: The **Map Number** shown below should be used when placing map orders; the **Community Number** shown above should be used on insurance applications for the subject community.

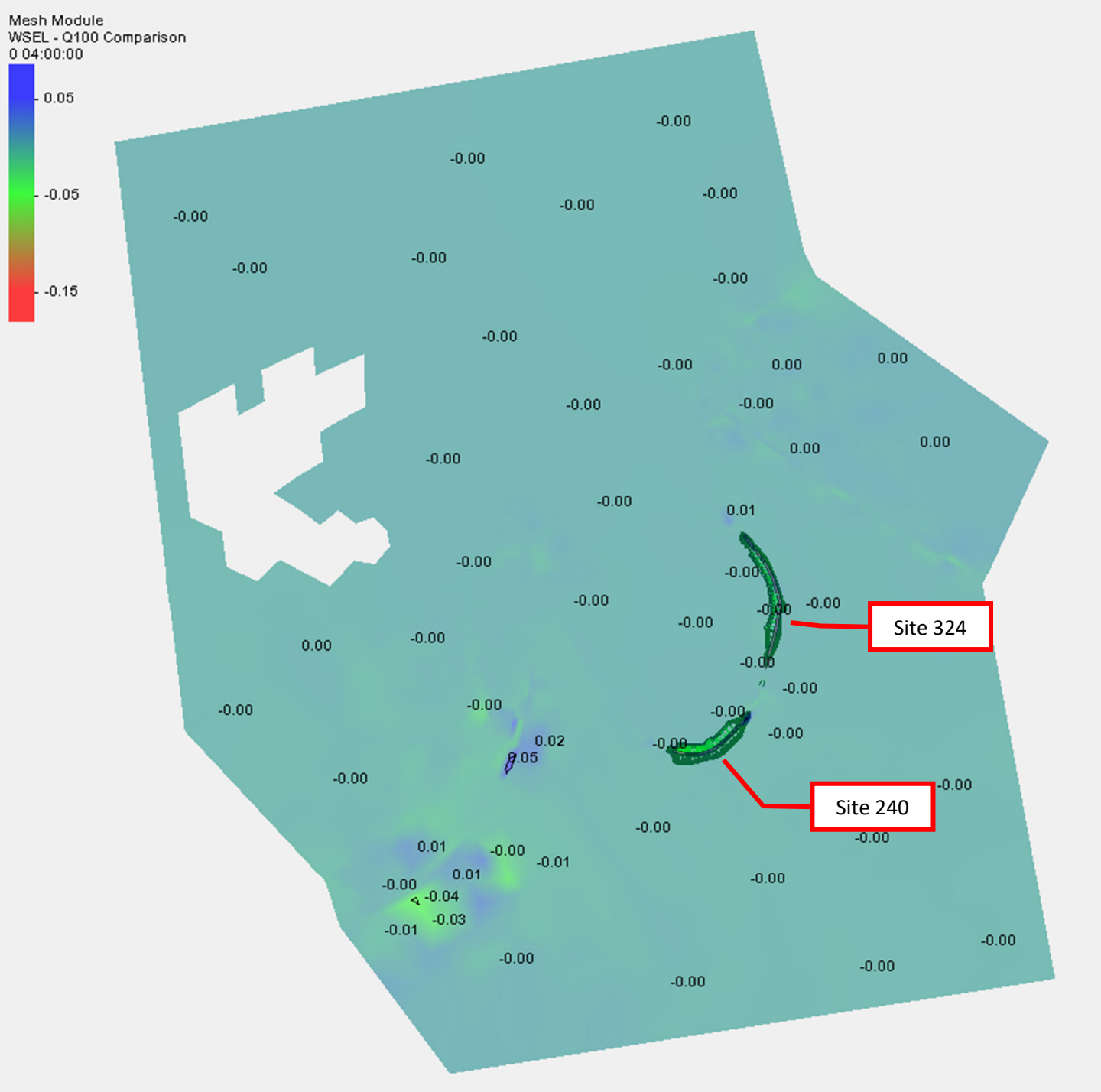
MAP NUMBER
36071C0417E

EFFECTIVE DATE
AUGUST 3, 2009

Federal Emergency Management Agency

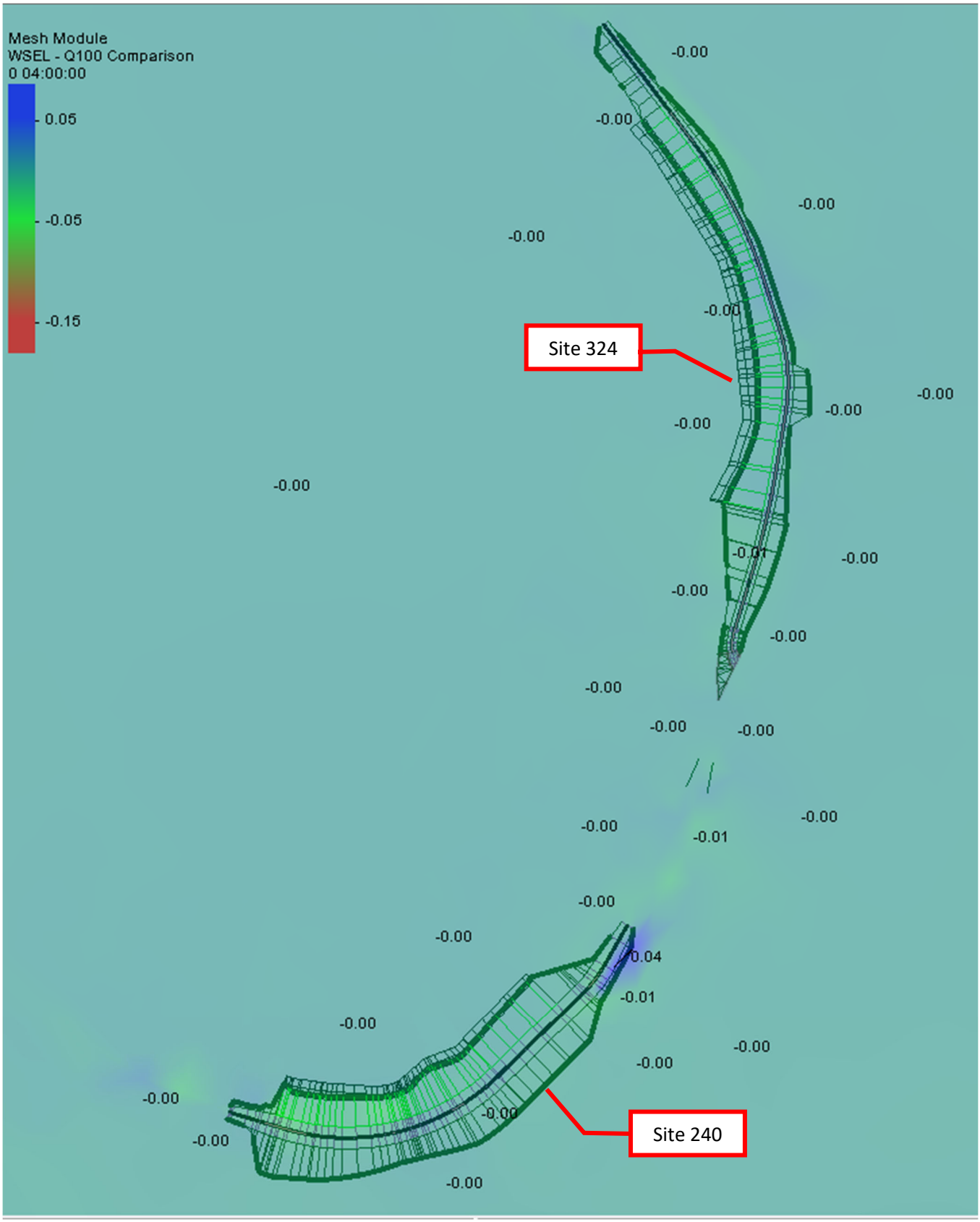
Attachment 21: NY WARI 2019-1(1)

SMS Plan View – Q100 WSEL Difference: Comparison of existing versus proposed Q100 WSELs across the SMS model study area.



No Scale

SMS Plan View – Q100 WSEL Difference: Comparison of existing and proposed Q100 water surface elevations (WSELs) at the project site.



No Scale